WHY VEIL? RELIGIOUS HEADSCARVES AND THE PUBLIC ROLE OF WOMEN*

Updated Version

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Abstract

We show that new economic opportunities that compel women to abandon their domestic roles have driven the adoption of religious veiling. We measure the prevalence of veiling among young women across Indonesia's districts for more than two decades by human-coding around a quarter million photographs attached to Indonesia's public high school yearbooks. We exploit exogenous variations generated by the interaction between international demand for Indonesia's products and sectoral and gender composition of local industries to show that the relationship is causal. This study suggests that veiling represents an effort by young women to reconcile their desire to join formal labor markets and the prevailing social norms in society.

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Economic development has delivered unprecedented opportunities for women, including new, appealing, and lucrative jobs, especially for those who are skilled or highly educated. This fact strongly characterizes the Indonesian labor market in the past decades (Schaner & Das, 2016). However, this new type of employment typically requires the workers to leave their homes to engage in the workplace over an extended period. In societies where women are designated as the primary bulwark of public morality and the traditional family institutions, taking this type of jobs may be costly for one's personal and social image, despite the economic benefits it offers.

In light of this dilemma, some women may devise strategies to lower the cost of participating in this type of employment using culturally compatible tools. Religious veiling appears to be one of these tools. Particularly effective in the context of Indonesia's highly religious society, some women adopt veiling in order to benefit from new economic opportunities but at the same time they can protect their dignity and reputation in society.

This study aims at documenting this phenomenon. In particular, we find that the participation of young women in the formal labor market has driven the adoption of veiling. The plot in figure 1 motivates this conjecture: Within the 22 year period of observation, the fraction of young females adopting religious veiling has evolved hand-in-hand with the fraction of young females participating in formal employment. This is in contrast, for instance, with the participation of young females in informal occupations.

We use an original method to measure the prevalence of veiling among young women by directly observing the portrait of students attached to Indonesia's public high school registers. The main data in this paper is based on human-coding of around a quarter million of these photographs. These data enables us to trace the evolution of veiling across districts for more than two decades. This new measure sets apart this study from previous research which mostly relies on surveys. Survey data on this social practice might not only suffer from reporting and social-desirability bias but also is not systematically available across space and time. To our knowledge, there is no comparable data on veiling of this scope available to date and the data collected for this study fills this void.

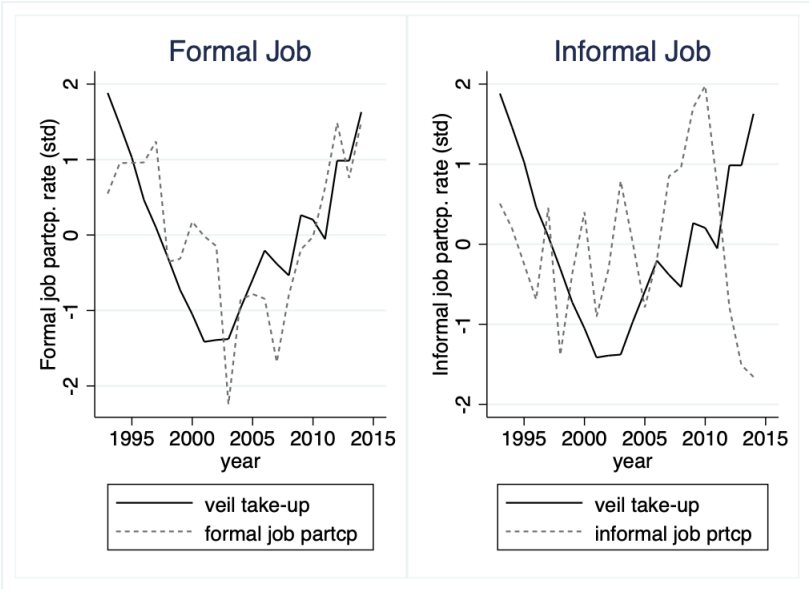


FIGURE 1 – (CO-)EVOLUTION OF VEILING AND FEMALE PARTICIPATION IN DIFFERENT TYPES OF EMPLOYMENT

Note: The graph shows the evolution of veiling by high-school female pupils against the evolution of formal and informal work participation by young females (age 20-24 years old) aggregated from all sample districts. All measures in this graph are detrended, to partial out the secular time trend, and standardized, to have a mean of zero and a standard deviation of one, hence the values are comparable across different measures.

To estimate the causal effect of economic opportunities on veiling, we build a Bartik instrumental variable exploiting exogenous variation generated by the interaction between international demand for Indonesia's products and sectoral and gender composition of local industries. This instrument creates a plausibly exogenous shock in the availability of economic opportunities for females by assigning a higher dosage of "treatment" in districts that specialize in female-intensive industries as opposed to districts that specialize in male-intensive industries. Hence, it allows the identification of the causal effect of economic opportunities on veiling.

Using this method, we find that young female participation in formal occupations has a positive, significant effect on veiling. In particular, every percentage point increase in the formal job participation by young females has caused increases in veiling by about 2.5 percentage points. This effect is equivalent to about one eighth of the mean value of veiling in the whole sample. Importantly, the estimates are robust to controlling for district

fixed effects, district-specific time trends, as well as additional covariates at the school, district, and national levels.

We explore several possible mechanisms. The evidence is consistent with a model in which veil is used as a negotiation device that enables young women to take employment outside of their homes but also helps to protect their respectability (Carvalho, 2013).[1] First, a number of independent qualitative studies as well as our own analyses using two different surveys confirm that education positively predicts the likelihood of women wearing a veil. Our interpretation is that more educated women are those who pays a higher cost of not joining the labor market (by wasting their educational investment), therefore they are more likely to veil to facilitate their access to the labor market. Second, while formal employment has a positive and strong association with veiling, participation in informal jobs or family enterprises are not associated with veiling. This implies that it is not just any job, but only jobs that require public engagement by women–hence create a threat to women's personal and social image–that drive the adoption of the veil. Third, we find that the effect of economic shocks on veiling is weaker as veiling becomes more common. This suggests that the veil might indeed has served as a signalling device since its signalling value dwindled as the practice is increasingly adopted.

A plausible alternative story is that the relationship between economic opportunities for women and veiling over time is explained by a concomitant rise in Muslim religious piety. The evidence suggests, however, that this does not seem to be the best explanation: First, we show that the effect of economic opportunities on veiling does not change with the inclusion of vote shares of Islamist parties–a measure of outward expression of religiosity. Second, drawing information from the Indonesian Family Life Survey (IFLS), we show that there is no significant association (reduced forms) between the instrument used in this study and three different measures of religiosity.

We provide evidence that the effects of economic opportunities may not be best understood as a cultural backlash, for instance, due to the pressure

[1] In this model, veil serves two function: First, it plays the role of a commitment device that helps women to keep the value they believe in. Second, it provides a signal to the community about the type of its adopters, hence protecting the adopters from the stigma related to working outside of her domestic compound (Carvalho, 2013).

from parents/family. Using nationally representative survey data, we find that around 9 out of every 10 veiled women declare that they adopt the veil due to their own desire as opposed to being requested by parents or spouse. Moreover, the women who adopt the veil tend to be those who take advantage of economic opportunities themselves, not those who stay at home. This implies that veiling may not be best understood as an act of protest by conservative women who are unhappy with the fact that more and more women leave their homes to work.

We rule out an alternative account that veiling is driven by personal income effects on the taste for fashion: we find no correlation between participation in informal job and veiling. This implies that improvements in income per-se do not induce veiling. We also find that economic shocks that are more consequential for male rather than female workers do not have similar effects on veiling rate. This latter evidence rules out an alternative mechanism where veiling is driven by a general increase in household income.

We address a concern that the main results may be driven by the changes in the composition of student bodies over time. First, we control for female high-school enrollment rate. This exercise does not affect the magnitude or the significance of the main estimates. Second, we do not find a correlation between female high school enrollment rate and veiling, which indicates that veiling by high schoolers is not a simple consequence of an ex-ante self-selection of girls from religious families into high schools as a response to economic shocks.

In the robustness analysis, we show that the coefficients in the main analysis do not change with the exclusion of outliers–schools that have 100 percent of students with veil, which may also indicate that the school mandates the students to veil. The results are also robust to the exclusion of observations with higher variance, as well as the exclusion of observations from a single year or a single district (Jackknife resampling estimation). We also find the results to be robust to alternative measurement of the instruments and different timing of economic shocks.

This study speaks to a broad range of literature in social science. First, the theoretical framework in this study is based on previous work on veiling by Carvalho (2013) and Patel (2012). Using cross-country survey data, Ak-

soy and Gambetta (2016) attempt to empirically test the theory, but they do not find support for the prediction of the model among less religious women. Our study, however, is the first to relate veiling practice with the transformation of female labor markets in developing countries. Apart from contributing original data, this study advances beyond the previously mentioned work by establishing a causal effect of new economic opportunities on veiling by using credible identification strategies and by providing evidence that rule out several alternative mechanisms. The logic of veiling as facilitating female's access to public has also been suggested in Abdelgadir and Fouka (2020). They show that a heardscarf ban is strongly associated with lower educational attainment for Muslim girls in France.

Second, our study contributes to the literature on the evolution of female participation in the economy e.g. Boserup (1986), Goldin (1990), and Fernández (2013) who highlight the role of culture and social norms, a perspective we share. In particular, this study is closely related to the literature on social norms and female labor force participation in Muslim-majority societies, e.g. Bursztyn, González, and Yanagizawa-Drott (2020). We contribute to this strand of literature by showing how the existing social norms adapt to new economic forces to produce a new set of norms. In particular, we show how a new norm of veiling is a result of economic modernization which, at the same time is being shaped by the local culture of gender relations.

The paper also contributes to the debate on the relationship between religion and economic development e.g. (Bazzi, Hilmy, & Marx, 2020; Bazzi, Koehler-Derrick, & Marx, 2019; Bénabou, Ticchi, & Vindigni, 2015; Binzel & Carvalho, 2017; Guiso, Sapienza, & Zingales, 2003; Squicciarini, 2019). In particular, our study offers a re-examination of the *secularization hypothesis* which associates economic development with decreased religiosity.[2] Our contribution to this debate is twofold: first, by focusing on the relationship between female economic outcomes and Muslim women's religious

[2]For instance, using analysis based on survey data, Norris and Inglehart (2004) and Voas and Crocket (2005) show a declining pattern of religiosity in Western Europe. This hypothesis was largely accepted until it was challenged by scholars such as Iannaccone, Stark, and others (Iyer, 2016). Iannaccone (1998) shows that religiosity in the US, arguably one of the most developed countries is much higher than in many less-developed nations. Moreover, Stark (1999) argues that despite rapid modernization, there is no demonstrable long-term decrease of religiosity in Europe.

practice, the study targets the heart of widespread curiosity about the position of women within Islamic culture. Second, the study speaks directly to the literature in this field by providing evidence suggesting that economic modernization does not always go hand in hand with secularization and the marginalization of religion.

We discuss the socio-economic background of the study in section I. The conceptual framework which outlines the theoretical reasoning of the article is elaborated in section II. Measurement of veiling and other variables is discussed in section III. We then elaborate on the empirical framework, the construction of the instrument, as well as the main results in section IV. We explore several plausible mechanisms in section V. Discussion on robustness checks is presented in section VI, while section VII concludes.

I Contextual Background

Indonesia provides an excellent ground to study the interaction between religious culture, gender norms, and economic development. First, the country currently hosts over 230 million Muslims, more than any other nation on earth. Moreover, Indonesia is arguably one of the most religious nations in the world.[3] Second, despite enjoying relatively higher autonomy compared to women in the neighboring region of South and East Asia, Indonesian womanhood in the past decades was shaped significantly by the ideology propagated by the New Order regime. Through regular women meetings (PKK-Pemberdayaan Kesejahteraan Keluarga /Family Welfare Empowerment program), the regime emphasized and glorified the domestic role of women and assigned them the role of the guardians of the traditional family institution.[4]

[3]The 4th and 5th waves of the World Value Survey (WVS) reveal that over 94 percent stated that religion is "very important" in life. In 2008, a survey by Pew Research Center shows that 80 percent of Indonesian Muslims perform five daily prayers, higher than other Muslim countries covered in the survey, including Egypt, Jordan, Pakistan, and Turkey. Using the Indonesian Family Life Survey (IFLS), Masuda and Yudhistira (2020) shows that the proportion of Muslims who only eat *halal* food in Indonesia is high and stable across periods at over 95 percent.

[4]These politically-driven gender norms still have strong impacts in many parts of Indonesia today. A UNDP survey in 2010 revealed that 94 percent thought that women should not work outside the home without prior permission from their husbands. The same survey showed that 95 percent thought that men should be leaders of households.

At the same time, the country has charted a robust economic growth in the past decades and since 1999, Indonesia has transformed from a military autocracy into one of the most vibrant, fledging democracy in Asia. Clearly, this development has delivered swift social and economic changes. These include rapid expansion of universal education (including for girls), reduction in total fertility rate, and a transformation of female participation in the economy.

In particular, there has been a process of "formalization" of female involvement in the labor force in the past decades. Two parallel phenomena accompany this process: first, younger women participate more in the formal sector of the economy (wage employment), especially in urban areas. At the same time, younger women in rural areas opted out of informal, unpaid employment (Schaner & Das, 2016). This trend is explained by the increasing access of young, highly educated female workers to lucrative and appealing jobs. Schaner and Das (2016) also documented that the wage gap between male and female workers had shrunk; in 1990 the median woman earned just 57 percent of what a man earned, but this figure had increased to 84 percent by 2011.

A The practice of veiling in Indonesia

While Indonesian Muslims consider religion as very important in their life, veiling by Muslim women was not a deeply rooted cultural practice.[5] This practice, especially since the 1970s, is arguably "new" since it differs significantly from veiling of the previous era. First, in terms of style, the veil of the previous generation, especially in Java, was commonly made from somewhat translucent fabrics which formed a drape that covers the head but still show the neck and parts of hair and ears. Meanwhile, the new veil is more enveloping, mainly made from an opaque material and

Another poll by CSIS shows that 86.3 percent agreed that "women's primary responsibility was to take care of the household" and 83.6 percent thought that women should not work at night (Robinson, 2018). A recent survey about gender roles in the Indonesian household reveals that only eight percent think that earning a living is part of the ideal role of a wife and 70 percent agree that doing household chores are the wife's responsibility (Jakpat Survey Report, 2020).

[5]In contrast, veiling has a long tradition among women who inhabited the area we call the Middle East today. It was established among Jewish and Christian women before (and after) the birth of Islam in the 7th century CE (Amer, 2014)

covering hair, neck, and ears while only showing the face (Brenner, 1996; Smith-Hefner, 2007). Second, while veils in the past tended to be adopted by few, commonly older, traditional women who lived in rural areas, the new veil was championed by young, educated, urban middle-class women (Smith-Hefner, 2007)[6].

These patterns are indeed confirmed by several recent studies. For instance, Fossati, Hui, and Negara (2017) find the proportion of female Muslim who answer "yes" when asked if they wear a headscarf is higher among the highly educated (94.5 percent) as compared to the low educated (78.4 percent). Moreover, the same survey also shows that the prevalence of veil is higher among the high-income group (84.9 percent) as compared to the low-income group (75.6 percent). In another recent study that predicts who wears the veil in the area of greater Jakarta, Utomo, Reimondos, Peter, Utomo, and Hull (2018) shows that veil use is significantly correlated with higher education level. These data and studies seem to suggest that in the Indonesian context, veiling seems to signify neither lack of education nor low social economic status.

The Politics of the veil. During its reign, the New Order regime has radically changed its stance toward the veil. In the early 1980s, the regime perceived the veil as a symbol of political threat. Hence, it tended to curb veiling, for instance, by issuing a decree that regulated student uniforms in high school.[7]. These rules induced many parents to disapprove of veil adoption by their daughters. Until the 1990s, veiling was a marginal practice which in fact, tended to be identified with an extreme form of religious piety by the general public.

However, president Suharto changed the course of his political strategy in the early 1990s. While consistently banning formal political forces based on Islamic ideology, he attempted to co-opt various Muslim groups by showing more friendly gestures to the Muslim community. For instance, he issued a regulation that lifted the ban on the use of headscarves in pub-

[6]This observation may seem puzzling given the widespread narrative about religious veiling, where it is mainly perceived as either a symbol of female's subjugation or as an expression of religious extremism. A literary critic, Said (1979), contends that this view was originally established during the colonial period as part of the "orientalist" narrative in the West. It was then gaining leverage in the aftermath of 9/11 terrorist attack.

[7]the decree SK 052/C/Kep/D.82 was issued on March 17 1982 (Johari, 2020)

lic high schools in 1991.[8] Following this change in regulation, the stigma attached to the veil began to fade and its popularity rose. The veil became even more widespread, popular among university and high school students, and also started to be adopted by older generations.

II Conceptual Framework

In order to understand how the emergence of new economic opportunities for women translate into veiling, we first need to understand the changes in terms of types of employment that accompany economic development over time. It is commonly observed that, as the economy develops and becomes more sophisticated, newly available jobs tend to be mentally-intensive as oppose to physically-intensive (Jayachandran, 2020)[9], hence highly suitable for women. Moreover, this process has also been typically accompanied by the shifting of production sites from households to factories or offices. Hence, workers need to leave their home to perform their job.

At the same time, development has also advanced female education, hence allowing younger generations of women to seize these economic opportunities. As a consequence, more and more young and educated women aspire to join formal labor markets and to do so they must abandon their domestic compound in order to pursue this aspiration.[10]

Unfortunately, social norms rarely keep up with the pace of economic transformation. As a result, women taking new types of employment may suffer from personal and social stigma. The extent of these costs may vary in different cultural contexts, but the common sources are discussed here.

[8] The decree number is SK.No.100/C/Kep/D/1991. In the same year, he also paid pilgrimage to Mecca and following his return, changed his official name to *Haji Mohammad Suharto*.

[9] The typical transformation involves the shift from traditional sectors, such as agriculture, fisheries and forestry, into manufacturing industries and then into services. The types of employment in the earlier phase are typically more physically demanding (farming, fishing, collecting forest harvests). When the economy is more developed, types of jobs in manufacturing and service industries are more mentally intensive (tailoring, fine goods crafting, accounting, administrative services).

[10] A similar line of argument is originally proposed by Becker (1981). He contends that newly available economic opportunities for women have changed the opportunity cost of staying at home, induced women to work, and consequently to have fewer children. This has led to a well-documented trend of declining fertility in the West in the 1970s.

First, women who take up formal employment may suffer from personal costs (guilt) because they perceive that working for mothers may lead to negative outcomes for children intellectual and emotional development (Fernández, 2013); or that it could jeopardize the stability of family life.[11] Second, taking a formal, high-paying job may signal ambition and autonomy[12], female attributes that often perceived negatively in the marriage market (Bursztyn, Fujiwara, & Pallais, 2017). Third, as the economy develops and lower-income families become more affluent, working women may suggest the inability of the husband or father to provide for the family (Goldin, 1990), hence create social stigma. Last but not least, taking formal employment is often considered "dirty"(Jayachandran, 2020) since it means that women have to mingle at the work place with males who are not family members.[13]

Given this dilemma, some women may attempt to adopt culturally-compatible solutions in order to lower the cost of joining the formal labor market. In the context of a religious Muslim-majority society like Indonesia, a veil appears to be one of these tools. There are at least two ways in which the veil could lower the personal and social cost associated with taking employment outside of the homes. First, at the personal level, the veil serves as an anchor (a *commitment device* in Carvalho's (2013) language) for the set of beliefs and norms that the women highly values. The veil binds sincere believers via their own statement of modesty, thus it helps them avoid conducts (or even thoughts) not in line with religious morality and values (Bénabou & Tirole, 2011; Carvalho, 2013). Second, the veil signals (albeit somewhat noisily) to the public the type of the girl who adopts it–even if she is not sincere (Carvalho, 2013; Patel, 2012). In particular, if the type signalled–e.g. feminine, rule-abiding, faithful–is desirable according to the prevailing norms, then this would help compensating for the possible stigma attached to working women. Consequently, *as the econ-*

[11]In the Indonesian context specifically, the women observed in this study have grown up in a regime that glorify women as the guardians of public morality and the institution of the family, a view propagated by the New Order Era from the early 1980s to late 1990s.

[12]In order to secure this kind of jobs, one often needs to pursue very high level of education, as well as additional skill trainings.

[13]Sometimes, they may even need to commute using crowded public transportation, which makes them susceptible to harrasment.

omy develops and there are more economic opportunities for women, more of them adopt the veil as they attempt to reconcile their desire to benefit from these opportunities, but at the same time want to avoid the negative personal and social image this act may generate.

It is important to note, however, that negative social attitudes toward working women are not unique to Muslim societies, but have also been a feature of the evolution of female labor participation in the West. The negative stigma documented by Goldin (1990), Fernández (2013), and Bursztyn et al. (2017) refers to the experience of the United States. What is interesting here is that the cultural context shapes how the conflict between social norms and the public role of women is resolved. In this case, the veil seems to be a functioning negotiation tool compatible with the setting of Muslim majority societies like Indonesia.

III Data Source and Descriptive Statistics

Veil. Measuring a cultural practice such as veiling is not a simple task. Previous studies rely on surveys to elicit information about veiling by directly asking female respondents whether they wear a veil. Alternatively, veiling is measured by simply observing if the respondent is wearing it during the interview. There are several issues associated with these methods. First, they might suffer from measurement error. For instance, if the surveyors are female themselves or if they conduct the survey at home, then a respondent who regularly wears a veil in public may not put the headscarf on during the survey.[14] Second, the traditional method may also suffer from reporting bias. For instance, if the surveyor is perceived by the respondents as having a certain degree of religious commitment (e.g. if the surveyor is wearing a headscarf herself), then this may result in biases in respondent answers.[15] Another major limitation of currently available data is that no source systematically traces the evolution of this practice across regions over time. Most of the surveys are ad-hoc and conducted at different times by different institutions.

[14]Commonly, headscarves are only be worn outside the home, where interaction with males who are not part of family members is pertinent.

[15]This phenomenon has documented, for instance, in Blaydes and Gillum (2013).

To overcome these limitations, we measure the headscarf take-up by directly observing the portrait of female students included in public high-school register books. We essentially measure revealed preferences for veiling, backed by official records of the subject's personal details. Public high schools in Indonesia are typically required to keep around 30 years records of student details in large register books where 3x4 CM picture of each student is attached.

Most of the pupils are between 16-19 years old and the pictures are taken mainly for the purpose of documentation for the national final examination. For this reason, the pictures are typically taken towards the end of the second year of high school or early in the final year, when most of the students may have already had an idea about what to do after graduation, e.g. to continue education, to work, etc.[16]

The sample of schools in this study is selected by randomly choosing schools in randomly chosen districts. In particular, the sampling is performed in two stages: First, 49 districts are randomly chosen from the sample frame of all 267 districts located in the two islands, Java and Sumatera[17]. The number of districts sampled in each province is proportional to the population in the provinces so that the sample is more representative in this aspect; hence, provinces with larger population have a higher number of districts included in the sample. In table 23 of appendix, one could observe the characteristics of sampling and non-sampling districts. Apart from population size, there is no systematic difference between the sampling and non-sampling group.

The second stage involves randomly choosing two schools out of the sample frame of all public high schools located in the district selected in

[16]There could be a concern that these pictures do not represent the behavior of the student in daily life. For instance, they may sometimes wear the veil and sometimes not or they might strategically use (or not use) the veil during the photo session. We argue, however, that this is not likely since Indonesian public schools require the pupils to wear school uniforms, which means they have a fixed set of clothing to wear on each day of the week. Moreover, the portraits are meant for identification purposes (like passport or ID card) in the national examination, so the picture should best represent the subject in their daily appearance.

[17]The original plan was to sample 50 districts, but one district dropped out of the sample due to challenging logistical problems. This number is chosen based on the limit of bugdet available for this project. The sampling is conducted in only two islands to maximize the efficiency of data collection. According to Statistics Indonesia, in 2010, 78.7 percent of the Indonesian population lives in these two islands.

the first stage. The sample frame excludes public madrasah (public Islamic schools) because all female pupils in public madrasahs wear a veil, so no variation/information is gained from these school. Similarly, private schools are also excluded from the sample frame because they are mostly religious schools, e.g. Catholic schools–where the veil is very rare–or Islamic schools–where all female pupils wear a headscarf, so no valuable information is gained from these schools. The enumerators are given a list of schools that have been randomly ordered and visit the schools based on this order. The procedure for data collection is reported in the appendix D. This section also contains the list of sample districts, as well as the geographic distribution of districts in the study.

Unless missing, the photographs of female (and male) students attached are of high quality in most books so we could clearly identify whether the student is wearing a veil or not and we could code veiling status by simply observing these pictures. Based on this information, we construct the measure of veiling rate, which is defined as the fraction of female pupils who wear a veil in a given school in a given graduation year. The evolution of veil take-up by district is presented in figure 2. The graph shows considerable variation across districts not only regarding the starting point of the upward trend but also in terms of the shape of the evolution. Some districts showed an initial increase in the early 1990s, but some others started late, after the year 2000. Some districts do not even start at all. Finally, in some districts, the prevalence of the practice goes up to 100 percent, while in others, the increase stops before reaching half of the population. One of the contributions of this paper is to explain part of this variation as a function of economic opportunities.

Labor force information. The data on female (and male) participation in the economy are sourced from the Annual National Labor Force Survey, the SAKERNAS. This survey is carried out by Statistik Indonesia and is rich with information related to working, schooling, and other activities performed by respondents representative of the population aged 10 and above. It also records information related to the characteristics of the job performed, e.g. the hours worked, if it is paid, information on personal characteristics of respondents, e.g. age, education level, and marital status, as well as respondents' location and characteristics, e.g. whether the

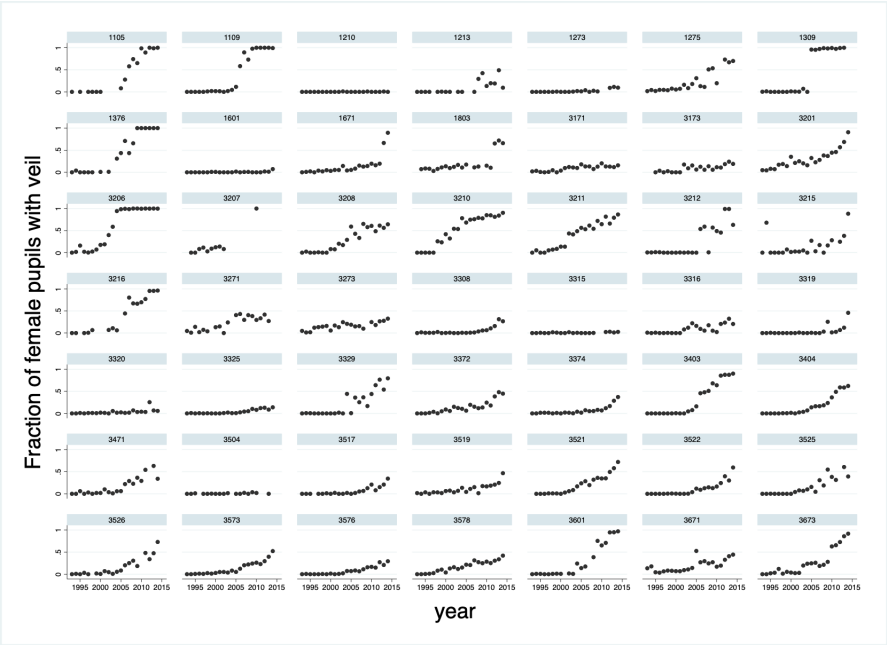


FIGURE 2 – Proportion of female students with veil, by district

Note: The figure displays the evolution of the veiling trend, separated by district. Each dot in the figure represents the fraction of female students who wear a veil in a given district, in a given year.

respondent lives in an urban or rural area. This source is used to construct the main variables of interest, including participation rate by female (and male) population in both formal and informal employment.

Formal employment in this paper is defined as that categorized as "worker/employee", while informal employment encompasses the rest of work categories, including family/unpaid workers, self-employed (with or without assistance), and casual labor (in agriculture and non-agriculture). This is a similar categorization to the one used by Schaner and Das (2016) in their ADB-comissioned technical report. This definition may not precisely capture the formality of the job as workers may also work in informal business. However, for the purpose of this paper, this differentiation is sufficient as it defines the likelihood of protection under the labor law, (in)flexibility in working time and place as well as the integration in the formal sector (Schaner & Das, 2016).

District sectoral and gender composition. These variables are used to construct the instrument and are calculated using the information from the Medium and Large Manufacturing census. This census is performed

annually by Statistik Indonesia and covers the universe of firms with at least 20 employees. It provides rich information about industrial establishments including the composition of industrial input: such as the number and gender composition of labor; electricity and fuel used; as well as the detailed composition of products/industrial output categorized by standardized classification (which could be matched to the ISIC–International Standard Industrial Classification). Using the information in this survey, we calculate the industrial composition of a district as well as the gender composition of workers employed in a given industrial sector.

International demand for Indonesian products. We use the information on international trade values to proxy the international demand for Indonesia's products sourced from the UN Comtrade. To construct the instrumental variables, the trade values by standardized industrial classification (nomenclatures) are then matched with the data from the industrial census previously discussed.

In table 1 of appendix we report the summary statistics of the main variables of interest. The mean value of veil take-up is around 20 percent in the whole sample with a pretty large standard deviation of around 29 percent.[18]The fraction of the female population (age 20-24) who participate in the formal sector is 21 percent on average, while the average participation in the informal sector is around 13 percent. These three variables are quite dispersed; they exhibit considerable variation across districts and time. Similarly, the export shock is also highly dispersed around its mean value.

Panel B of the table present the components of the instrumental variable: The average real value of exports in the sample is around 7 times its value in 1990 and is very dispersed with a standard deviation of 12. The average share of a single sector is around 7 percent. This implies that the districts covered have a pretty diversified industrial composition. On average, female workers constitute around 38 percent of labor in all units

[18]Please note that this mean is calculated for all observation, meaning that it computes across different years. The mean value across districts in the early years tends to be lower than the mean in the later years. Importantly, this measure is in line with the national survey data: the mean value of veiling at the national level in 2014 based on our data collection from high school is around 65 percent, which is very close to the value estimated by a recent survey in 2016 at around 70 percent, e.g. Fossati et al. (2017) or the data from Jakpat polling firm in the same year, which will be discussed in the following sections.

of observation covered in the analysis; the lowest fraction of female labor is employed in iron and stell basic industries (5 percent) and the highest fraction is employed in tobacco manufactures (79 percent).

Independent surveys and other controls. We also perform some analyses using independent survey data from Pew Research Institute, a local marketing polls company (Jakpat polling firm); as well as the Indonesia Family Life Survey (IFLS). In the main analysis, we add some additional control variables using data from various other sources, including the World Bank and KPU(Komisi Pemilihan Umum/National Election Committee). The complete list is reported in appendix C.

IV Estimating Equations and Empirical Results

A OLS Estimates

To explore the the relationship between veiling and young female participation in formal occupations, we estimate the following equation as a starting point:

$$Veil_{sdt} = \beta_1 \; FemJob_{dt} + \mathbf{X'}_{s,d,n;t}\mathbf{\Gamma} + \delta_d + \tau_d T + \epsilon_{sdt} \tag{1}$$

$Veil_{sdt}$ denotes the fraction of female students who wear a veil at school s, in district d, and year t. The main explanatory variable, $FemJob_{dt}$, denotes the fraction of young women (aged 20-24 years) who work in the formal sector in district d, and year t. We do not use the employment measure for the younger age group (16-19 years old) for an obvious reason: this group is mostly still in high school, so the employment measure of this group will be biased, especially in the current years. Importantly, since the picture is taken toward the end of high school years, the employment measure we use is the best proxy of job market opportunities for women captured in the pictures.

$\mathbf{X'}_{s,d,n;t}$ denotes a vector of control variables at different levels of observation, the details of which are discussed below, when the results are presented. These controls include variables at the school ($\mathbf{X'}_{st}$), the district ($\mathbf{X'}_{dt}$), and the national level ($\mathbf{X'}_{nt}$). The inclusion of this set of

variables aims at controlling for time and unit varying information that may also explain veiling. In order to control for non-time-varying characteristics specific to the districts, we also include district fixed effects, δ_d. Meanwhile $\tau_d T$ is meant to capture district-specific linear time trends; and ϵ_{sdt} is an error term.

Given this specification, what we exploit is the variation within districts over time, after partialing out the drift from the increasing time trend of veil take-up specific to the district. This specification is motivated by the observation from figure 2 where we observe that different districts experience a different level of (increasing) time trend throughout the observation period. Moreover, since the treatment is applied at the district level, we cluster the standard errors for all regression specifications at the district level.

Since the treatment, i.e. the export shock, is most likely to take effect in multiple times period and the effect is most likely to be heterogeneous based on the position along which the observation lies on the veiling curve across different points in time, we do not simultaneously control for time fixed effects. As demonstrated by de Chaisemartin and D'Haultfœuille (2020), Goodman-Bacon (2020), and Imai and Kim (2020) among others, in this setting, using a two-way (unit and time) fixed effect may not represent the best estimation strategy for causal inference.[19]

The main analysis in this study is using a dataset with school-district-year as a unit of analysis, containing 93 schools located in 49 districts within the timespan of 1993 - 2014.[20] The estimates for equation 1 is presented in table 1. The coefficient β_1 is positive, highly significant, and stable across different model specifications. From the second column onwards, we also control for a vector of confounders at different levels of observations. At the

[19]Using time and unit fixed effect in a panel data often is seen as equivalent to difference-in differences estimator. In a setting where the treatment is applied at various point in time and where there is heterogeneity in effect, using this technique could lead to a wrong average treatment effect estimate due to possible negative weighting. The studies cited show that the two-way fixed-effects specification for estimating ATE in a panel data structure is best applied in the simplest setting of panel data (two groups, two time period)

[20]The reasoning for this period is twofold: first, the information on gender labor composition from the Manufacturing database is only available from 1993 onwards. This starting year is also good to avoid the censoring effect of the government regulation of public high school uniform which was lifted only in 1991.

TABLE 1 – OLS ESTIMATES: VEIL AND FORMAL JOB PARTICIPATION

	Veil				
Formal job participation	0.172	0.172	0.159	0.166	0.151
	(0.062)	(0.062)	(0.053)	(0.061)	(0.052)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.761	0.770	0.771	0.771	0.772

Note: The table reports OLS estimates of equation 1. The unit of analysis is school-district-year. *Veil* is the fraction of female students who wear a veil, *Formal job participation* is the fraction of female population age 20-24 who works in the formal sector. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties, and female high-school enrollment rate; National-level control includes economic growth rate. All specifications include district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level

school level, we might be worried that veiling is related to the ratio of girls and boys in the classroom. This ratio may affect the "competition" among girls for available boys (or vice versa), and thus might affect how female students dress to influence how they are perceived by their peers at school. When the competition is high, we may expect that female students may want to stand out among their peers, so they may be putting on (or off) veil in order to achieve this goal. Moreover, the size of the student cohorts might also affect veiling. A larger student body might reduce the visibility of individual students within the school, hence it might affect the decision to veil. The geographic location of schools may also affect veiling because this defines the climate in the area and hence influences how people dress to adjust to the weather. Controlling for this set of variables at the school level barely changes the magnitude of the effect of interest.

Several controls at the district level are also included: First, the fraction of population living in urban areas may affect veiling through the level of

social surveillance experienced by the women in the study. In rural areas, people may know the members of the community better, so women in rural areas may experience a higher degree of scrutiny over their actions. This may affect their decision regarding attire and to adopt the veil. Moreover, we control for young male employment because male and female employment rates may move together. As a consequence, if male employment is the true driver of veiling, then the interpretation of the effect would be different, as discussed in the next section, where alternative mechanisms are evaluated. By including this variable in the specification, the effect we observe from female employment is free from the confounding effect of male employment. In addition, we also control for high-school enrollment rate by female population of the relevant age groups. This variable might capture the effect of socio-economic background and composition of the student body in different district-years. For instance, when schools are more accessible, a higher attendance rate means that the socio-economic composition might be closer to the actual socio-economic characteristics of the population. Meanwhile, this might not have been the case in the past, when high school might only be accessible to daughters of families from high socio-economic background. Another control included at the district level is the share of votes for Islamist parties. This variable is meant to capture changing social preferences for outward religious expression that may simultaneously affect the taste of population regarding veiling. Finally, at the national level, we include the national economic growth rate, which controls for possible time-specific confounders from economic growth that affect all district in a similar fashion. For instance, a national economic crisis or years of recovery could affect the year-specific standard of living of the population in all districts. The inclusion of these possible confounders alters neither the magnitude nor the precision of the estimates in any substantial manner.

B Shift-share Instrumental Variable

The estimate of β_1 in equation 1 might be biased due to some unobserved characteristics that caused districts with higher female job opportunities to also have a higher prevalence of veil-take up. For instance, districts with

more vibrant economic activity may also be inhabited by more religious people, so we simultaneously observe more veiling.

To address this problem, we build a Bartik-style instrumental variable (IV) to instrument female participation in the formal sector. This strategy has been widely used in economics and social sciences, for instance by Autor, Dorn, and Hanson (2013) to estimate the effect of imports from China to local US labor markets; Card (2009) to understand the impact of immigration on local labor markets in the US, and Tabellini (2020) who uses this type of IV to identify the political and cultural explanation of the backlash against immigration. The essence of the identification strategy is to use plausibly exogenous shocks coming from international sources to predict their effect on local labor markets. In line with the spirit of this strategy, we are using international demand shocks for Indonesia's products, which are weighted by district industrial and gender labor composition to predict female labor demand or more precisely female participation in formal employment.

The logic is the following: assume there is an increase in the demand for tobacco products in international markets. This induces tobacco industries to hire more labor. Importantly, tobacco industries systematically hire more women than men. As a result, districts with a large share of tobacco industry experience higher demand shocks than districts that specialize in other commodities. Since international demand is by and large exogenous, the changes it induces in female labor demand are plausibly also exogenous.

In particular, the instrument, *export shock*, is calculated as the summation of normalized real values of exports for different industries weighted by their historical gender and industrial composition:

$$export\ shock_{dt} = \sum_{k=1}^{K} value_{k,t} \underbrace{\frac{L_{k,d,t=1993}}{L_{d,t=1993}}}_{sector\ \ share_{kd}} \underbrace{\frac{L_{f,k,t=1993}}{L_{k,t=1993}}}_{\text{female score k}} \tag{2}$$

The subscript notation d is the identifier for the district, t is for time, k is the sector in the economy and K is the number of sectors in the economy. The first component, $value_{kt}$ is the normalized real value of

export commodities produced by sector k at time t.[21] Since this variable is measured at the national level, it varies across years but is constant across districts within a given year. The second component, $sector\ share_{kd}$ is the historical share of sector k in district d in 1993 (the base year). This number captures the relative importance of industry k in a given district, hence, its value varies from district to district but is constant for each district over time. The third component, $female\ score_k$ is an index which captures the relative importance of female as opposed to male labor force in a given industry k. For instance, textile and tobacco manufacture tend to have a higher share (more females); meanwhile, steel and heavy machinery manufacture tend to have a lower share (fewer females). It is calculated as the fraction of female workers in the industry at the base year, 1993. This number is common across districts and time but varies across sectors[22].

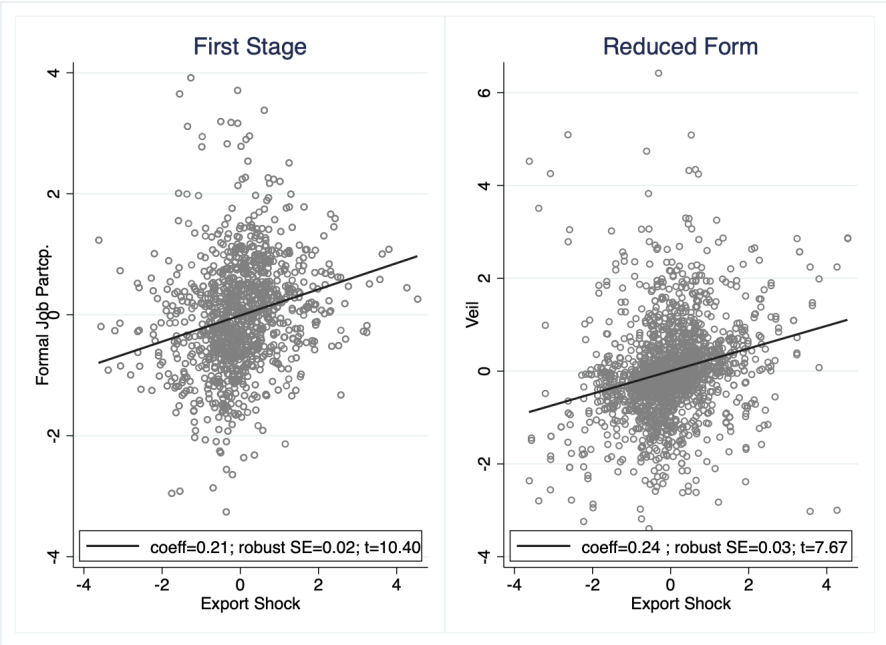
Given its structure, *export shock* affects different districts in different years based on the fluctuation in the international demand which is amplified by the gender characteristics of the industry and the share of that industry in a given district. The summary statistics of the components of the instrument are presented in panel B of table 1. The step-by-step calculation process of this instrumental variable is discussed further in appendix C.

In figure 3, we show the scatter plot of the instrument and the two variables of interests. First, *export shock* has a clear positive and significant relationship with formal job participation for females (first-stage). Similarly, export shock is also positively and significantly associated with veil take-up (reduced form). Importantly, these relationships are obtained after partialing out time-invariant district characteristics as well as district-specific time trends.

[21]This value denotes real increase, so it accounts for the inflation rate and it is normalized to have a value of 1 in 1990.

[22]We do not use district-specific female score because it may be endogenous to the characteristics of the labor force of the district; national level scores are more exogenous. In any case, we perform a robustness check with district-specfic female score in section 20 of appendix and this exercise does not change the results.

FIGURE 3 – THE FIRST STAGE AND THE REDUCED FORMS



Note: This figure displays relevant correlations for the first stage on the left hand panel and of the reduced form on the right hand panel. The variables are residualized (of district fixed effects and district-specific time trends) and standardized (with a mean of zero and a standard deviation of one) to make the scales comparable across different variables

C The causal effect of economic opportunities on veiling

To identify the causal effect of shocks in economic opportunity on veiling, we estimate the following model with two-stage least squares (2SLS):

$$FemJob_{dt} = \alpha\ export_shock_{dt} + \mathbf{X}'_{s;d;t}\ \mathbf{\Omega} + \eta_d + \ \theta_d T + \varepsilon_{dt} \quad (3)$$

$$Veil_{sdt} = \beta_2\ \widehat{FemJob}_{dt} + \mathbf{X}'_{s;d;t}\ \mathbf{\Gamma} + \delta_d + \tau_d T + \epsilon_{sdt} \quad (4)$$

export shock denotes the instrumental variable introduced earlier, and the rest of the notation corresponds to equation 1. Our parameter of interest here is β_2: a positive value indicates that the increase in economic opportunity for women increases the prevalence of veil use.

We run the 2SLS regression following model 4 and report the estimates in table 2. From the table, the reader may observe that the first stage relationship is strong and significant: when the weighted average of export value increases to twice its value in 1990, formal job participation is ex-

TABLE 2 – 2SLS ESTIMATES: VEIL AND FORMAL JOB PARTICIPATION

Panel A. REDUCED FORMS ESTIMATES					
			Veil		
Export shock	0.062	0.061	0.059	0.060	0.058
	(0.012)	(0.012)	(0.013)	(0.012)	(0.014)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.772	0.781	0.781	0.781	0.782

Panel B. SECOND-STAGE 2SLS ESTIMATES					
			Veil		
Formal job participation	2.324	2.313	2.680	2.312	2.682
	(0.690)	(0.685)	(0.872)	(0.710)	(0.933)

Panel C. FIRST-STAGE ESTIMATES					
			Formal job participation		
Export shock	0.026	0.026	0.022	0.026	0.021
	(0.006)	(0.006)	(0.005)	(0.006)	(0.005)
K-P Wald F-statistics	23.14	22.83	18.93	22.14	16.84

Note: The table reports reduced forms and 2SLS estimates of equation 3 and 4. *Veil* is the fraction of female students who wear a veil, *Formal job participation* is the fraction of female population age 20-24 who works in the formal sector, *Export shock* is the instrument. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties and female high-school enrollment rate, National-level control includes economic growth rate. Panel B and Panel C also include the same set of additional controls (School, District, and National levels) as Panel A. All specifications include district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level

pected to increase by about 2.6 percentage points. This value is a little less than 10 percent of the average value of female employment in the sample. Meanwhile, the reduced form regression indicates that an increase in weighted average export value to twice its value in 1990 is associated with about 6.2 percentage points increase in veiling. Moreover, both the first stage and the reduced form coefficients are significantly different from zero at the one percent level. Importantly, the reported Kleibergen-Paap Wald F-Statistics are larger than 10 in all specifications, attesting to the strength of the first stage regression.

Finally, the 2SLS estimates indicates that on average, every percentage point increase in formal employment participation by female is associated with about 2.3 - 2.7 percentage point increase in veiling and these estimates are highly significantly different from zero at 99% confidence interval. To put these numbers in perspective, this effect is equivalent to about 11 - 13 percent of the average veiling in the sample.[23]

Comparison to OLS estimates. Although similar qualitatively, the magnitude of the coefficient resulting from the two methods differs. These differences could be attributed to several factors: First, *omitted variable bias*. As previously discussed in section B, the estimate of β_1 is likely to be biased due to unobserved characteristics that are strongly correlated to both female participation in the formal sector and veiling rate. One such possible characteristic is the level of religiosity. One may expect that religiosity positively correlates to veiling. At the same time, religiosity may negatively be associated with female labor force participation. Assuming that these associations are valid at the individual level and that they on average carry through at the district level, their combination will likely lead to downward bias in β_1, i.e. since districts with more religious people are more likely to have lower female employment but at the same time are more likely to have higher veiling rate. Then, the direct OLS estimates of veiling on female employment will be biased toward zero.

[23]We could also gauge the extent of this magnitude by looking at table 1. From this table we can observe that although the mean value of both veiling and female formal employment is about the same (20 percent), the standard deviation of veiling is twice as much as of female employment. Therefore, since the breadth of the veiling variable in this sample is about twice as wide as the range of female employment variable, while they have the same unit, the effect of about 2.5 percentage point is very well within the reasonable estimates.

Second, *measurement error*. At the theoretical level, the phenomenon we are trying to capture is female labor demand. This is likely to be only noisily measured by female participation in formal jobs. This variable, in fact, is the equilibrium of the demand and the supply of female labor. Supply could be affected, for instance, by the progressive women movement for female emancipation. However, this progressive movement may be negatively correlated to veiling. Again, this would push the estimate of interest downward.

Third, *local treatment effect*. The first stage regression analysis reported in supplementary figure 2 shows that there is large heterogeneity among districts. The effect may be diluted in the OLS regression since in the OLS estimator, we average over all observations. Meanwhile, the estimate from the 2SLS regression is mostly applied to compliers (local treatment effect), which is the effect of job opportunities that are affected by export-driven "booms" and "busts" in areas that specialize in female-intensive commodities.

D Validity of the instrument.

Recent work on SSIV can mainly be categorized into two broad camps. The first camp argue that the validity of SSIV mainly hinges on the assumption that there is exogenous variation coming from the common shock received by the district while allowing the shares to be endogenous, e.g. Adão, Kolesár, and Morales (2019) and Borusyak, Hull, and Jaravel (2018). Meanwhile, the second camp contends that the validity of SSIV should depend on the exogeneity of the shares of the sectors, e.g. Goldsmith-Pinkham, Sorkin, and Swift (2020). The earlier scenario is very likely to be satisfied in the setting of this study since we are not using a common shock generated by national production that serves the local markets. What we are exploiting instead, are the shocks that are coming from international markets. Hence, this source of variation is plausibly exogenous since it is less likely to be affected by local market conditions.[24]

Meanwhile, the scenario in Goldsmith-Pinkham et al. (2020) is more

[24] As a robustness, in section A of appendix, we also perform an analysis using global imports of commodities, rather than the value of export.

likely, for instance, when the design emphasizes a two-industry setting or emphasize shocks to specific industries as the central element of the design. However, in this paper, we use more than two dozen industrial sectors, a third of which has more than half female share in the labor force (please refer to appendix 22 for the complete list). Hence, the analysis of this study is less likely to depend on a few specific industries.

Moreover, since we fix the industrial share at the base year, the plausibility of the design assumes that the export shock should be independent of the initial features of the district. This assumption could be violated if there are some characteristics of the districts which make them persistently receive higher (or lower) shocks in the long run due to initial features of the districts. To gauge the validity of this assumption, we perform several checks:

1. We add several possible confounders at the district level in the main analysis and show that the estimate of interest is robust. The idea is to show that the relationship between the instrument and the outcome of interest does not depend on the variation coming from these pertinent socio-economic features of the district. We report the results in table 2 from the second column onwards.

2. We show that future export shocks cannot predict past veil take-up. The idea is the following: If there is some systematic correlation between past veil take-up and distant future economic shocks, then one may suspect that the instrument is not independent of the initial condition of the districts, including the local industrial sectoral composition, because it is fixed at the base year. For this purpose, we run both the reduced forms and the 2SLS estimates for the first and the last specification of table 2, using lead-instrument. In particular, the lead-instrument is calculated as follows:

$$export_shock_{d,t+10} = \sum_{k=1}^{K} value_{kt+10} \underbrace{\frac{L_{kd,t=1993}}{L_{d,t=1993}}}_{sector_share_{kd}} \underbrace{\frac{L_{fk,t=1993}}{L_{k,t=1993}}}_{\text{female score_k}} \tag{5}$$

Here, $value_{kt+10}$ denotes the 10-years-lead of normalized real values of national exports, while the rest of the notation follow equation 2.

In essence, this lead instrument captures the future economic shocks that the district would receive in 10 years.[25]

We report this analysis in table 3 of the appendix. The future export shock is not correlated to past veil take-up both in the reduced form and in 2SLS regression, despite the fact that the first stage still holds. The coefficients are tiny and unstable across different specifications and importantly, they are not statistically different from zero.

3. We show that economic shocks are balanced on several past characteristics of the district. The intuition of this exercise is to show that the future economic shock is not correlated to the initial characteristics of the district. We report the results in supplementary table 4 and show that there is no systematic correlation between future shocks and the initial characteristics of the district, such as the fraction of population who live in urban areas, female high school attendance rate, male job participation, votes for Islamist parties, gender ratio in the population, etc.

   These results strengthen the argument that the instrument is indeed plausibly exogenous.

V Mechanisms

Thus far, we have presented evidence which suggest that an exogenous shock in the availability of female formal job opportunities has a causal impact on the adoption of veiling. In this section, we attempt to gauge possible mechanisms and provide several pieces of evidence regarding alternative explanations.

[25]The main idea of this exercise is to see if there is an unobserved characteristics of the district at the base year that persist in the long run so that the future shocks are correlated to these past characteristics. To test this, we need to use a long enough time span (e.g. 10 years or more), with the purpose of ensuring that these dependencies (if any) do not carry through in the distant future. Using a shorter span of the lead instrument would not be ideal since companies are typically using short or medium term plans for their activities, e.g. over 3 or 5 year ranges there might be stickiness, since firms may need to follow the prescribed plan within this time span.

A Veil as a negotiation device

As elaborated in section II, the veil might have been used as a negotiation device in dealing with the dilemma regarding the public and domestic role of women in the context of a rapidly modernizing, religious society. In this section, we triangulate several pieces of evidence that compel us to believe that this is perhaps the most likely underlying mechanism. First, several studies confirm that the new trend of veiling in contemporary Indonesia was initiated and championed by young, urban, educated middle class women (Brenner, 1996; Fossati et al., 2017; Smith-Hefner, 2007; Utomo et al., 2018). We also run simple logistics regressions using two different sources, one collected by a local polling firm and another by an American-based polling firm. The analysis reveals that there is indeed a positive correlation between education attainment and the likelihood of women wearing a veil in Indonesia. In table 5, the coefficient of education is positive and highly significant in both datasets, the values of the coefficients are in fact comparable to each other.[26]

Despite the observational nature of the evidence, the confirmation of this trend by multiple studies and analyses suggests that this pattern seem to fit well with the veiling model in Carvalho (2013). Consistent with the model, these women face the strongest incentive to veil because a) They are those who would pay the highest opportunity cost of not joining the labor force, i.e. wasting their educational investment and foregoing access to higher paying jobs; and b) Since they are likely to be higher paid, they will likely receive higher stigma from society.

Second, we find that veiling is only sensitive to the type of jobs that may pose a threat to women's respectability. An underlying component of the mechanism discussed in Carvalho (2013) and Patel (2012) concerns respectability while accessing the public space. In the context of paid em-

[26]Moreover, the reader may want to have a peek at table 11 of appendix to observe veiling rates for women with different level of education using the data from the local polling company. Moreover, table 2 of appendix present the comparison of veiled and non-veiled group of Pew Research respondents on various socio-demographic characteristics. The two groups of women are very similar except in two attributes: first, veiled women tend to be more educated and two, they are less likely to be married–which may also imply higher financial independence. These differences are even stronger for the younger cohort of respondents.

ployment, wearing a veil signals to the community that the women who earn from a lucrative job are indeed earning from a "legitimate" source, i.e. that do not compromise their dignity. An extreme demonstration of this function is documented in an ethnographic study by Lindquist (2004) who studied young female workers in the context of the rapidly developing industrialized area of Batam, Indonesia. He argues that young female workers in the manufacturing companies in the area don a veil in order to establish the moral boundaries within their environment: to differentiate themselves from commercial sex workers. Stigma for young working women who are accused to work as prostitutes has also been documented via ethnographic study from an export processing zone in South Sulawesi (Silvey, 2000).

To test this reasoning, we run an analysis showing the association between veiling and different types of female jobs. In particular, here we are contrasting formal employment and informal employment. In the Indonesian context, informal work by women is typically done around the home, allowing women to combine domestic and earning income activities[27] (Robinson, 2009). Moreover, our data also listed a category of family worker, which mainly refers to the type of jobs within one's family enterprise, e.g. family business, and often unpaid.[28] The idea here is to check if this latter type of job may not be sensitive to veiling because a) The scope of job does not necessarily require women to leave their domestic compound and/or b) For "family worker' especially, work often is unpaid, so there is no concern for the "undignified" economic exchange involved as there is no monetary exhange in the first place.

For this purpose, we compare the regression estimates of veiling on

[27]One may argue that some types of informal work may still induce women to leave the house, e.g. selling food in nearby market. However, what is important here is simply the fact that formal employment is more likely to require the worker to leave their homes. To validate this assumption, in table 14, we use the data from the 2014 Sakernas to show that formal jobs is more likely to be associated with using means of transportation to go to work (as oppose to going on foot) and formal job is also associated with longer commuting distance to work. Moreover, we also use family worker category as a comparison. For this category, we can be more certain that this type of work is perform in the vicinity of the home or other family members

[28]The Sakernas has different coding for this category in different years, category 5 in early years (before 2001) and category 7 in the later years, but the label of job category is defined as "Family/upaid workers."

female participation in the formal sector and in the informal sector, as well as the family worker category. The results are presented in table 6. In the bottom panel, we replace formal job participation with its more exogenous measure, export shock. Across different types of female occupations, there is a contrast between the estimates for formal and informal occupations. The coefficients on informal and family workers participation are very small and none of them is significant; in some specifications they actually have negative sign. On the other hand, the estimates for formal job participation (and export shock) are consistently positive, stable, and highly significant across different specifications. We interpret this contrast as a consequence of the different types of jobs that need to be performed in terms of exchange for money and/or contact with the public.

Finally, we perform a heterogenity analysis which suggest that the veil may indeed have been used for signalling. The idea is that the value of signalling will be highest when the veil is visible; consequently, as veiling becomes more common, its value as a signalling device will be depleted. Hence, we expect that the effect of economic shocks on veiling will be smaller as the adoption of veil in the previous period is higher and it may not even have a significant effect any longer when veiled women are the majority in the population. We run the reduced form regressions to understand the effect of economic shocks on veiling at different level of previous veil adoption using the following model:

$$\begin{aligned} Veil_{sdt} =& \alpha_1 \; export_shock_{dt} + \alpha_2 \; Veil_{sd,t-1} + \alpha_3 \; export_shock_{dt} * Veil_{sd,t-1} \\ &+ \mathbf{X}'_{s;d;t} \; \mathbf{\Omega} + \eta_d + \; \theta_d T + \varepsilon_{dt} \end{aligned} \tag{6}$$

In graph 4 we present the marginal effects of export shock on veiling, while the details of regression estimates are presented in table 15 of appendix. Under this modelling, the effect of economic shocks is weaker as there are more women already adopting the veil. Interestingly, export shocks do not seem to be significantly affecting veiling when the fraction of veiled students in the previous period is about 60 percent. In other words, when veiling becomes a majority, adopting it might no longer be an effective signalling device in the community.

To sum up, the analyses above suggest a compelling case for interepret-

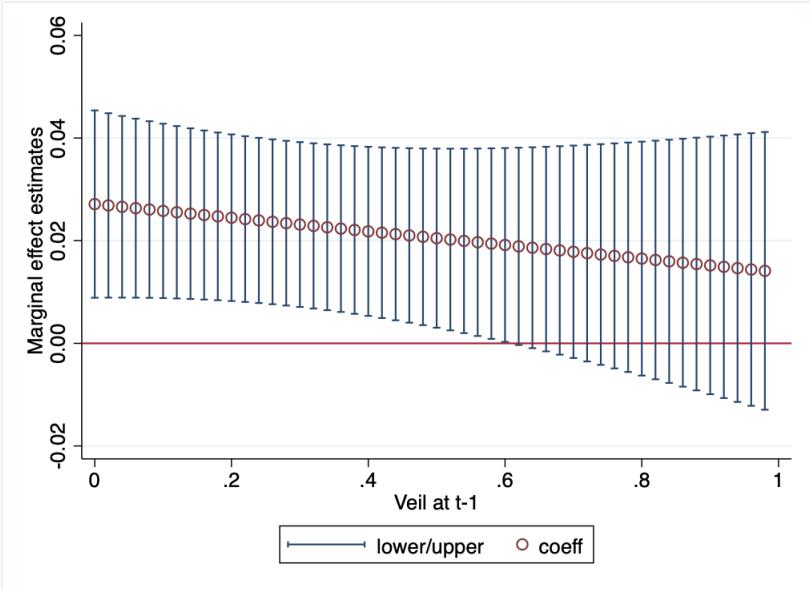


FIGURE 4 – HETEROGENEITY EFFECTS BY VEILING AT *T-1*

(a) *Note:* The figure displays the estimates of the marginal effects of export shocks on veiling in the reduced forms following equation 6. The outcome variable is veiling at time t, while the marginal effect of export shocks is estimated at different level of veiling at t-1. The hollow dot represent the estimated marginal effect, while vertical lines represent the 95% confidence interval.

ing veiling as a negotiation device: First, the socio-demographic characteristics of veil adopters documented both in survey data and qualitative studies suggest that veiling is taken up mostly by women who have to deal with the challenges of arbitrating their beliefs or social norms and the new opportunity presented by economic development. Second, we show that veiling is only sensitive to the type of jobs that may pose a threat to women's perceived respectability and that it does not seem to react to increased participation in jobs where this threat is less likely to present. Finally, signalling may indeed be a crucial ingredient of the mechanism explaining the association between veiling and economic shocks, since the effect is diluted as veiling becomes more common.

B Concomitant increase in religiosity

Apart from the possible mechanism discussed above, another possibility is that economic shocks may also correlate with the level of religiosity, for instance, as documented by Chen (2010). Moreover, religiosity might be expected to correlate with veiling. Hence, the increase in veiling may actu-

ally be driven by a concomitant rise in Islamic piety, not by the availability of new economic opportunities.

There are several pieces of evidence which suggest that this may not be the best explanation for the pattern we just documented. First, there is suggestive evidence, presented in table 2 of appendix that even at a personal level, religiosity is not a good predictor of whether the women respondent wear a veil or not. There is no significant difference in the number of daily prayers or support for Islamist parties among the veil and non-veil group of respondents. Second, we include the vote share of Islamist parties in a number of specifications of the main analyses and the coefficient of interest does not change with the inclusion of this variable[29]. Third, we perform an additional analysis drawing the information from IFLS (Indonesian Family Life Survey) to understand if there is a positive association between the instrument used in the main analysis and various measures of religiosity.[30] The three measures are a self-reported level of religiosity, the number of prayers performed in a day, and the importance of religion and religiosity in influencing preferences for political candidates.

The results are reported in table 7. The correlations between the instrument and each of the three different measures of religiosity are not statistically significant. The sign is even negative for the last measure of religious politics. These results undermine the alternative mechanism through an increase in religiosity. This is not saying, however, that religion has nothing to do with the adoption of veiling. The underlying logic of why veiling might be effective as a signalling device is largely based on beliefs attached to religious morality or values. What these results simply say is that religiosity does not interfere with the effect of economic opportunity on veiling we estimated above.

C Cultural backlash

As there are more economic opportunities that compel more women to abandon their traditional role, there may be a cultural backlash from con-

[29]We may think of preference for Islamist parties as a measure of preference for outward expression of religiosity, which might be correlated to preference for veiling

[30]These measures are not available in the earlier waves, hence the analysis is performed using the data from the last two waves.

servative groups in society. Below, we discuss some evidence related to this mechanism.

Parental Pressures. It might be the case that the women sampled in this study adopt the veil because they are still very young and therefore, highly influenced by the pressures to put on a veil either from their parents or their family. The evidence however, does not strongly support this reasoning. First, if this were true, we should find a significantly higher rate of veiling among younger women as opposed to older women. In table 10 we show data from a national survey that the veiling rate in younger age group is not significanly higher than in the older age group (if anything, it tends to be lower). Second, in table 8 of appendix, the same survey shows that the overwhelming majority of women respondents who wear a veil declares that they adopt the veil because of their own will despite alternative options of "requested by parents" or "requested by spouse" available in the survey.[31]

Protest by Conservative Women. A possible form of backlash might be manifested by conservative groups of women who purposely veil to express their opposition toward the fact that more women are leaving their homes to work. If this is the case, then we would expect that veil adopters are more likely to be those who are staying at home (because they despise women who abandon their traditional roles) and not those who are taking the economic opportunity themselves. However, there seem to be no strong evidence to support this reasoning. First, as previously shown in table 5, two different surveys confirm that education has a positive association with the probability of female respondents wearing a veil. This implies that veil adopters tend to be those who would pay the higher cost of staying at home.

Second, we run a logistic regression of the binary variable for whether the respondent wears a veil on different dummies for working status using the data from JAKPAT, a local polling companies. The result of this analysis is presented in table 9 of appendix. Relative to women who are working or going to school, women who stay at home are significantly

[31]There might be a concern that the answer is biased, for instance, because the respondents are under the surveillance of their parents or husbands when responding to this question. However, this should not be a concern because this survey is not conducted face-to-face, but using a mobile phone app where anonymity is expected and social desirability bias should be minimal.

less likely to wear a headscarf. These analyses show that this alternative mechanism does not seem to find strong support in empirical patterns.

D Compositional effect

An alternative mechanism that might explain the above pattern is a compositional effect. It has been documented by several recent studies, e.g. Heath and Mobarak (2015); Jensen (2012), that increasing economic opportunities will induce an increase in the demand for female education. As a result, girls who previously were staying at home, begin to attend high schools: If these girls are positively selected from conservative religious families, then what we observe is a compositional effect from having more girls from religious families attending public high school.

It is unfortunate that our data contain neither information about religiosity of the students nor of the students' family, hence, a direct evidence for this reasoning is difficult to obtain[32]. However, a number of analyses do not support this interpretation. First, in some specifications of the main analyses, we include the female high school enrollment rate and this inclusion does not change the main effect[33]. Second, in table 12 we show that veiling does not have a significantly positive corelation to female high school attendance. If it were true that the increase in veiling is simply the consequence of an ex-ante selection from religious families into high school, then we should see a correlation between veiling and female high school attendance rate, i.e. the higher the female high school attendance rate, the higher is the level of veiling.[34] The evidence, however, does not support this reasoning. Third, if this type of compositional effect is the primary

[32]The IFLS contains information about the educational history of respondents. Unfortunately, there are simply not enough observations in the data to construct a reliable measure of the fraction of public high school attendees who have an Islamic educational history, e.g. Islamic elementary or Islamic junior high school, at the cohort-district-year level and gender.

[33]Higher enrollment rate means that the composition of student body are closer to the socio-economic composition of the population. This means that the composition of students in terms of religiosity also change with the changes in enrollment rate.

[34]In contrast, the results on the table from the second column onwards includes the main variable of interest, i.e. female formal job participation and export shocks. The analysis shows that when put together, the coefficients of female HS enrollment rate are unstable and are not different from zero, while the main variables of interest are still positive and significantly different from zero across different specification.

driver of veiling, then we should not see the "conversion" into veiling among the older generation of women, since they are already out of high school. From the survey data presented in table 10 of appendix we can observe that veiling is also an observable phenomenon among the older cohort of women: it is unlikely that veiling among these groups of older women to be driven by self-selection into high school.

Another compositional effect may come from the fact that economic opportunities may compel high school pupils to drop out of their school to obtain the newly available jobs. As a consequence, if the students who stay are positively selected from conservative religious backgrounds, then our estimates may capture this compositional effect instead. However, this scenario is very unlikely in the Indonesian context: the data from Statistics Indonesia shows that the average national dropout rate for general high school attendance is very low, i.e. less than 2 percent. Moreover, the chance of obtaining formal employment for a high-school drop-out is really low for most formal jobs, even blue-collar occupations, like in manufacturing industries (Robinson, 2009). Therefore, we may rule out that our main results are driven by confounding of this type.

E Income effect on the taste for fashion

An alternative reading is that the positive effect of economic opportunities on veiling could simply be the consequence of income effects on the taste for fashion in the female population. As the economy develops and the female population experiences improvement in income, they might start to think about expenditures beyond necessity, such as fashion. If the veil is considered a normal good, then its consumption increases as income increases. Alternatively, income effects may also be channeled through household income. If family income is pooled, and female subjects are permitted to spend from this pool, then the increase in veiling may be due to the household income effect, e.g. from increasing income of male family members.

To address this concern, we performed the following exercises. First, we add a measure of economic growth in some of the main specification: this inclusion does not significantly alter the coefficient of interest. This

exercise supports the interpretation that veiling does not seem to be driven by increasing general income or wealth. Second, analysis presented in table 6 shows that veiling has no correlation with informal job participation. This suggests that an increase in female's income per se (via informal employment, which typically does not require women to leave their domestic compound) does not necessarily induce adoption of the veil. Third, in some specifications in the main analyses, we also control for young male labor force participation and this inclusion does not change the main estimates. Moreover, in table 13 we show estimates of male labor force participation on veiling. Despite a relatively strong first stage effect, the second stage effect of male labor force participation on veiling, is not only much smaller, but also is not significantly different from zero. This is consistent with the interpretation that an income increase for male members of the household, which potential increases the disposable income for women, does not have a similar effect on veiling.

VI Robustness analyses

In this section, we summarize several robustness checks to support the validity of the analyses. The corresponding tables are in appendix.

Removing outliers. One may be concerned that some districts may mandate female pupils to wear headscarves, e.g. *sharia* law that requires public servants and female pupils in public schools to wear headscarves on Fridays. If these areas happen to also display high economic participation in the formal sector, we may pick up the effect of these regulations instead. To address this concern, in table 16, we rerun the analysis of table 2 removing the district-years that have one hundred percent of female pupils with a veil. This is a good proxy of the existence of mandatory veil regulation in the school/district, and also a way to trim off the outliers in our sample. The results are robust to the exclusion of these observations. In fact, the coefficient of interest is actually larger.

Removing observations with higher variance. Because in some districts we could only collect the data from a single high school, there is higher uncertainty on the veiling measure in these districts. To address this, we

perform the analysis excluding district-years observation with only one high school and report the results in table 17. The estimates are very similar to the ones in the main analysis.

Jackknife resampling estimation. We perform a Jackknife re-sampling procedure to make sure that the results reported above are not driven by any single district or any single year of observation. This procedure is performed by iteratively estimating the main model leaving one set of observations out of the sample (i.e., all observations from a given district or a given year). We present the results in figures 3 and 4 of the appendix. This analysis corroborates the main results: the coefficients are similar to the main results, stable, and systematically different from zero.

Timing. In the main model, we assume that the response to labor market shocks happened simultaneously[35]. However, there might be a delay in the responses to the economic shocks. As a robustness check, in table 19 of appendix we show that the relationship still holds with the lagged value of export shocks. However, the estimates become weaker and less significant as the lag increases.

Instrument with district-specific female score. The instrument used in the main analysis is using female score that is constant across districts, i.e. using national level scores. One may be concerned that there might be a strong heterogeneity coming from different levels of technological advancement by district which is overlooked when we are using national level female score index. As a robustness check, we also run the main 2SLS regression using a district-specific female score index and the results are presented in table 20. There is no significant change both in terms of the magnitude or the significance of the results.

Alternative IV with global imports. The IV used in the analysis is using the value of Indonesia's export as the main element of the common economic shock. As discussed in section D, the validity assumption requires that this element of the IV is exogenous. This is very likely to be satisfied given that we are exploiting variation from International markets. We run a robustness check by running the analysis using an alternative IV with global imports, instead of using the country's export value. The details

[35]Veiling is neither a very costly act nor taking time to do like building a school. Therefore, one can respond to the incentives as soon as the opportunity arrives

of the alternative IV as well as the results are reported in subsection A of section robustness checks of appendix. The results are robust to this alternative instrument.

VII Discussion and conclusions

Economic development has delivered unprecedented opportunities for women in the form of appealing and lucrative employment. However, prevailing social norms, especially those that govern the division of labor between men and women, inside and outside the house, may create a hurdle for women to seize these opportunities.

In the face of this dilemma, religious veiling may provide an effective, culturally compatible tool in the context of a highly religious society like Indonesia. It could serve as a negotiation tool that enables women to join the formal sector of the economy, but it also protects them from the possible personal and social stigma generated by taking employment outside the home, deviating from the prevailing norms.

In this paper, we document this phenomenon by collecting new, innovative data on veiling based on revealed preferences from photographs of public high school pupils. This method makes it possible to trace the evolution of veiling across districts in the country for more than two decades. We use a Bartik-style instrument to show that female participation in the formal sector has driven the adoption of veiling. The instrument takes advantage of the historical district industrial composition, the female labor composition of sectors, as well as plausibly exogenous shocks in international demand, and it allows us to identify the causal effect of shocks in economic opportunities for women on veiling. Moreover, we show that this effect is not likely to be the result of concomitant increases in religiosity, income effects on the taste for fashion, compositional effects of the student body, or a cultural backlash.

There remain open questions that our design cannot address directly. Our data cannot trace the relationship between veiling and life-cycle decisions of women. For instance, our data does not really allow us to see whether women with veil are more likely to delay marriage, have fewer children, or retain employment. Are they more likely to exit the labor market

once they have children or get married as they may be more committed to family life? Future studies may want to focus on understanding these decisions of women at later stages in life.

Nevertheless, the results in this paper may shed some light on the general observation of the increased popularity of veiling in Indonesia in particular and in other Muslim majority countries in general. We do not aim, however, at convincing the readers that female employment is the only driver of veiling among women in contemporary Muslim society. Indeed, there are multiple interpretation as well as motivations of why Muslim women veil. What we document in this paper is a phenomenon where in the context of highly religious, rapidly modernizing societies like Indonesia, economic development that creates new opportunities for women may have a causal effect on the adoption of the veil which might reflect a negotiation process between modernity and the prevailing social norms.

Finally, the results presented in this paper may also provide an alternative reading on the meanings and motivations of this religious practice and perhaps some reference in designing relevant policies in the advanced world, e.g. on assimilation or integration of minority groups. Importantly, cultural values and social norms might shape the process of economic development in ways that seem counter-intuitive at a first glance.

References

Abdelgadir, A., & Fouka, V. (2020). Political secularism and muslim integration in the west: Assessing the effects of the french headscarf ban. *American Political Science Review*, *114*(3), 707–723.

Adão, R., Kolesár, M., & Morales, E. (2019, 08). Shift-Share Designs: Theory and Inference. *The Quarterly Journal of Economics*, *134*(4), 1949-2010.

Aksoy, O., & Gambetta, D. (2016). Behind the veil: The strategic use of religious garb. *European Sociological Review*, *32*(6), 792-806.

Amer, S. (2014). *What is veiling.* The University of North Carolina Press.

Autor, D. H., Dorn, D., & Hanson, G. H. (2013, October). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, *103*(6), 2121-68.

Bazzi, S., Hilmy, M., & Marx, B. (2020, May). *Religion, education, and development* (Working Paper No. 27073). National Bureau of Economic Research. Retrieved from http://www.nber.org/papers/w27073 doi: 10.3386/w27073

Bazzi, S., Koehler-Derrick, G., & Marx, B. (2019, 12). The Institutional Foundations of Religious Politics: Evidence from Indonesia. *The Quarterly Journal of Economics*, *135*(2), 845-911.

Becker, G. (1981). *A treatise on the family*. Harvard University Press.

Bénabou, R., Ticchi, D., & Vindigni, A. (2015). Forbidden fruits: The political economy of science, religion and growth. *NBER Working Paper No. 21105*.

Bénabou, R., & Tirole, J. (2011, 05). Identity, morals, and taboos: Beliefs as assets. *The Quarterly Journal of Economics*, *126*(2), 805-855.

Binzel, C., & Carvalho, J.-P. (2017). Education, social mobility and religious movements: The islamic revival in egypt. *The Economic Journal*, *127*(607), 2553-2580.

Blaydes, L., & Gillum, R. (2013). Religiosity of interviewer effects: Assessing the impact of veiled enumerators on survey response in egypt. *Politics and Religion*, *6*(3), 459-482.

Borusyak, K., Hull, P., & Jaravel, X. (2018, September). *Quasi-experimental shift-share research designs* (Working Paper No. 24997). National Bureau of Economic Research.

Boserup, E. (1986). *Women's role in economic development*. Gower.

Brenner, S. (1996). Reconstructing self and society: Javanese muslim women and "the veil". *American Ethnologist*, *23*(4), 673-697.

Bursztyn, L., Fujiwara, T., & Pallais, A. (2017). 'acting wife': Marriage market incentives and labor market investments. *American Economic Review*, *107*(11), 3288-3319.

Bursztyn, L., González, A. L., & Yanagizawa-Drott, D. (2020, October). Misperceived social norms: Women working outside the home in saudi arabia. *American Economic Review*, *110*(10), 2997-3029.

Card, D. (2009, May). Immigration and inequality. *American Economic Review*, *99*(2), 1-21.

Card, D., Mas, A., & Rothstein, J. (2008, 02). Tipping and the dynamics of segregation. *The Quarterly Journal of Economics*, *123*(1), 177-218.

Carvalho, J. (2013). Veiling. *The Quarterly Journal of Economics*, *128*(1), 337-370.

Chen, D. L. (2010). Club goods and group identity: Evidence from islamic resurgence during the indonesian financial crisis. *Journal of Political Economy*, *118*(2), 300-354.

de Chaisemartin, C., & D'Haultfœuille, X. (2020, September). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, *110*(9), 2964-96.

Easterly, W. (2009, June). Empirics of strategic interdependence: The case of the racial tipping point. *The B E Journal of Macroeconomics*, *9*(25). doi: 10.2202/1935-1690.1492

Fernández, R. (2013). Cultural change as learning: The evolution of female labor force participation over a century. *American Economic Review*, *103*(1), 472–500.

Fossati, D., Hui, Y.-F., & Negara, S. D. (2017). *The indonesia national survey project: Economy, society and politics.* Institute of Southeast Asian Studies (ISEAS) - Yusof Ishak Institute.

Goldin, C. (1990). *Understanding the gender gap: an economic history of american women.* Oxford University Press.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020, August). Bartik instruments: What, when, why, and how. *American Economic Review*, *110*(8), 2586-2624.

Goodman-Bacon, A. (2020). Difference-in-differences with variation in treatment timing. *NBER Working Papers*, *w25018*.

Guiso, L., Sapienza, P., & Zingales, L. (2003). People's opium? religion and economic attitudes. *Journal of Monetary Economics*, *50*, 225-282.

Heath, R., & Mobarak, A. M. (2015, January). Manufacturing growth and the lives of bangladeshi women. *Journal of Development Economics*, *115*, 1-15.

Iannaccone, L. R. (1998). Introduction to the economics of religion. *Journal of Economic Literature*, *36*(3), 1465-95.

Imai, K., & Kim, I. S. (2020). On the use of two-way fixed effects regression models for causal inference with panel data. *Political Analysis*, 1–11.

Iyer, S. (2016). The new economics of religion. *Journal of Economic Literature*, *54*(2), 395-441.

Jayachandran, S. (2020). Social norms as a barrier to women's employment in developing countries. *NBER Working Paper Series*, *w27449*.

Jensen, R. (2012, March). Do labor market opportunities affect young women's work and family decisions? experimental evidence from india. *The Quarterly Journal of Economics*, *127*(2), 753-792.

Johari, H. (2020). Jilba terlarang di era orde baru. *Historia online*. Retrieved from https://historia.id/kultur/articles/jilbab-terlarang-di-era-orde-baru-6k4Xn

Lindquist, J. (2004). Veils and ecstasy: negotiating shame in the indonesian borderlands. *Ethnos*, *69*(4), 487-508.

Masuda, K., & Yudhistira, M. H. (2020). Does education secularize the islamic population? the effect of years of schooling on religiosity, voting, and plurarism in indonesia. *World Development*, *130*, 104915.

Norris, P., & Inglehart, R. (2004). *Sacred and secular: religion and politics worldwide*. Cambridge University Press.

Pan, J. (2015). Gender segregation in occupations: The role of tipping and social interaction. *Journal of Labor Economics*, *30*(2), 365-408.

Patel, D. S. (2012). Concealing to reveal: The informational role of islamic dress. *Rationality and Society*, *24*(3), 295-323.

Robinson, K. (2009). *Gender, islam and democracy in indonesia*. Routledge.

Robinson, K. (2018). Gender culture and politics in post-new order indonesia. In R. W. Hefner (Ed.), *Routledge handbook of contemporary indonesia*. Routledge.

Said, E. W. (1979). *Orientalism*. Vintage.

Schaner, S., & Das, S. (2016). Female labor force participation in asia: Indonesia country study. *ADB Economics Working Paper Series No.476*.

Silvey, R. M. (2000). Stigmatized spaces: gender and mobility under crisis in south sulawesi, indonesia. *Gender, Place and Culture*, *7*(2), 143-161.

Smith-Hefner, N. (2007). Javanese women and the veil in post-soeharto indonesia. *The Journal of Asian Studies*, *66*(2), 389-420.

Squicciarini, M. P. (2019). Devotion and development: Religiosity, education and economic progress in 19th-century france. *CEPR Working*

Paper DP13877.

Stark, R. (1999). Secularization, r.i.p. *Sociology of Religion*, *60*(3), 249-273.

Tabellini, M. (2020). Gifts of the immigrants, woes of the natives: Lessons from the age of mass migration. *The Review of Economic Studies*, *87*(1), 454–486.

Utomo, A., Reimondos, A., Peter, M., Utomo, I., & Hull, T. (2018). Who wears the *Hijab*? predictor of veiling in greater jakarta. *Review of Religious Research*, *60*(4), 477-501.

Voas, D., & Crocket, A. (2005). Religion in britain: neither believing nor belonging. *Sociology*, *39*(1), 11-28.

Appendix

A Supplementary tables and figures

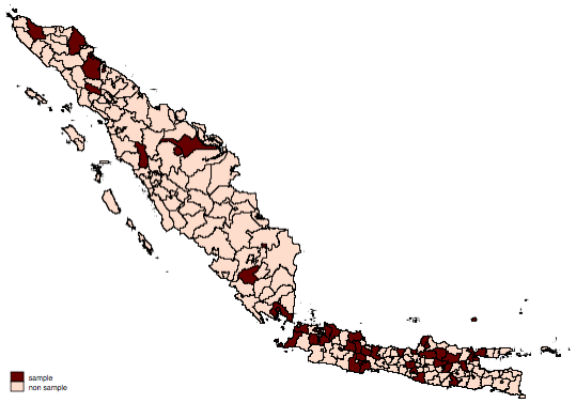


FIGURE 1 – GEOGRAPHIC DISTRIBUTION OF SAMPLE DISTRICS

Note: The figure displays the location of districts sampled spread over two islands: Java and Sumatera. Since the number of districts sample is proportional to the population in the province, more districts are sampled in Java (bottom right) than in Sumatera (top left)

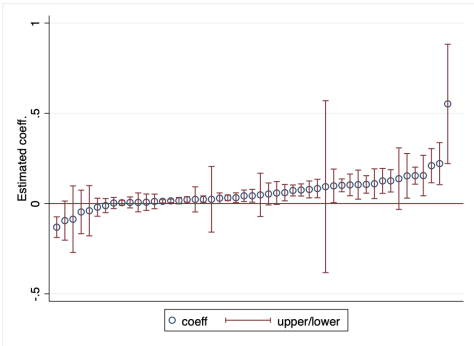


FIGURE 2 – FIRST-STAGE HETEROGENEITY EFFECT

Note: The figure display the heterogeneity of the coefficient of the first stage regressions. Each dot represents the coefficient of the OLS estimate (within 95% confidence interval) of female formal job participation on export shock by district

TABLE 1 – SUMMARY STATISTICS

VARIABLES	N	mean	sd	min	max
A. Main dataset					
Main Vars of interest					
Veil	1,687	0.204	0.293	0	1
Formal job partcp.	1,687	0.208	0.141	0	1
Export shock	1,687	1.991	1.487	0.168	11.44
Other outcomes					
Informal job participation	1,687	0.131	0.123	0	1
Family worker	1,687	0.0675	0.0962	0	1
Male job partcp.	1,687	0.605	0.159	0.100	1
Additional covariates					
Fraction urban	1,687	0.520	0.346	0	1
Islamist votes	1,687	17.38	7.873	4	35
Economic growth	1,687	4.446	4.266	-13.13	7.818
Female HS enrollment	1,687	0.532	0.162	0	1
Num. of student	1,687	262.5	102.5	28	792
Latitude	1,687	-5.421	3.681	-8.076	5.358
Longitude	1,687	107.8	4.480	95.94	112.8
Male/female ratio	1,687	0.932	0.951	0.00865	37
B. Components of the instrument					
Value of export	15,290	6.966	12.07	0.298	90.90
Sector share	15,290	0.0705	0.137	0.000298	1
Female score	15,290	0.381	0.195	0.0534	0.793
C. Religiosity measures					
Religiosity	13,654	2.819	0.623	1	4
Num. of Prayers	12,637	4.513	2.528	0	81
Relig. politics	13,338	6.076	1.682	0	8
D. Jakpat polling survey (2016)					
Veil	996	0.714	0.452	0	1
Education level	986	3.361	0.687	1	4
Working status	996	2.051	0.787	1	3
Age	996	24.57	5.768	14	56
Spending group	996	2.150	1.188	1	6
Socio-Economic Status	996	3.988	1.537	1	6
E. Pew survey (2012)					
Veil	662	0.225	0.418	0	1
Age	662	29.21	6.355	18	40
Education level	662	2.275	0.905	1	4
Income group	662	6.159	2.959	1	18

Note: The unit of observation in panel A is school-district-year, in panel B is industry-district-year, in panel C is individuals surveyed in IFLS wave 4 and 5, resident of districts included in the sample, in panel D and E is female respondents surveyed in the respective surveys

TABLE 2 – SOCIO ECONOMIC CHARACTERISTICS: VEIL VS. NON-VEIL

Characteristics	Age: 18 - 40			Age:18 - 30		
	No Veil	Veil	Difference	No Veil	Veil	Difference
age	29.06	29.74	-0.68	24.83	24.55	0.28
			(-0.59)			(0.48)
married	0.82	0.75	**0.07**	0.74	0.61	**0.13**
			(0.04)			(0.06)
education level	5.34	5.66	**-0.31**	5.47	6.02	**-.56**
			(0.17)			(0.22)
income level	6.14	6.21	-0.06	6.05	5.9	0.14
			(0.28)			(0.37)
num. of children	1.79	1.86	-0.06	1.4	1.5	-.09
			(0.12)			(.17)
access internet?	1.86	1.8	0.05	1.83	1.76	.07
			(0.03)			(.049)
access facebook?	1.08	1.04	0.04	1.02	1	.019
			(0.06)			(.033)
own cellphone?	2.01	2	0.01	2.02	2	.018
			(0.02)			(.038)
own landline?	1.97	1.97	0.003	1.98	1.97	.009
			(0.014)			(.017)
num. of daily prayer	7.18	7.21	-.035	7.12	7.15	-.033
			(0.138)			(0.19)
support Islamist party?	0.29	0.3	-.006	0.27	0.31	-.037
			(.0547)			(0.07)
urban	0.47	0.38	.085*	0.45	0.39	.062
			(.046)			(0.06)
Observation	513	149		307	80	

Note: The table presents socio-economic characteristics of respondents grouped by their veil status. The unit of analysis is individual and the data source is The Muslim World Survey, 2012 by Pew Research Institute. Standard error in parentheses. Veil in this survey is coded based on surveyors' observation about the repondents during the interview, hence the number of respondent with veil is most likely understated.

TABLE 3 – OLS REDUCED FORMS AND 2SLS WITH LEAD INSTRUMENT

	$Veil_{sdt}$			
	Reduced form		2SLS	
	(1)	(2)	(3)	(4)
Export shock$_{d,t+10}$	-0.005	-0.002		
	(0.012)	(0.012)		
Formal job partcp.$_{d,t+10}$			-0.287	-0.139
			(0.742)	(0.693)
School-level controls		✓		✓
District-level controls		✓		✓
National-level controls		✓		✓
Observations	788	788	788	788
R-squared	0.547	0.623	0.528	0.619
			First Stage	
			Formal job partcp.$_{d,t+10}$	
Export shock$_{d,t+10}$			0.0162	0.0158
			(0.008)	(0.007)

The unit of analysis is school-district-year. The first two columns report the reduced form regressions of veiling on 10 years lead instruments, while the last two columns present the 2SLS regressions of veiling on female participation in formal employment, instrumented by 10 years lead export shock. All specifications includes district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level.

TABLE 4 – BALANCE TESTS

	Share urban$_t$	Female HS enrollment$_t$	Male job$_t$	Islamist votes$_t$	Log population$_t$
Export shock$_{d,t+10}$	-0.001	-0.001	-0.006	-0.542	-0.044
	(0.012)	(0.013)	(0.014)	(0.733)	(0.017)
Observations	788	788	788	788	788
R-squared	0.934	0.705	0.636	0.547	0.907

	Share fem popul$_t$	Informal job$_t$	Family worker$_t$	Num of student$_t$	Male to female ratio$_t$
Export shock$_{d,t+10}$	0.012	0.000	0.002	-6.580	0.024
	(0.014)	(0.015)	(0.011)	(5.657)	(0.041)
Observations	788	788	788	788	788
R-squared	0.263	0.659	0.580	0.569	0.082

Note: The table displays the OLS regression of various characteristics of districts on 10 years lead of instruments. All regressions includes district fixed effects and district specific time trends and standard errors are clustered at the district level.

TABLE 5 – LOGIT REGRESSION OF VEIL ON EDUCATION ATTAINMENT

	Pew Survey		**Jakpat Survey**	
	age <41	age <31	age <41	age <31
	veil (yes=1)		veil (yes=1)	
education level	0.257	0.486	0.45	0.44
	(0.124)	(0.189)	(0.11)	(0.13)
Observations	658	385	970	859

Note: The raw data sources are surveys by *Pew Research Institute* in 2012 and Jakpat polling in 2016. The table reports logistics regression of whether the respondent is wearing a veil or not on education attainment. For Jakpat survey: SE is clustered at the district level and all regression is controlling for age and dummies for spending group (proxy for income). For Pew survey: all regression also control for age and dummies for income group. The coding for education category is rescaled so that they are comparable between different surveys.

TABLE 6 – VEILING AND DIFFERENT TYPES OF FEMALE JOBS

	Veil			
	(1)	(2)	(3)	(4)
Panel A. All informal jobs				
Informal job partcp.	0.015	-0.004	-0.007	-0.012
	(0.075)	(0.066)	(0.074)	(0.067)
Formal job partcp.	0.175	0.150		
	(0.061)	(0.052)		
Export shock			0.062	0.058
			(0.012)	(0.014)
Panel B. Family workers				
Family worker	0.019	0.023	-0.022	-0.013
	(0.047)	(0.047)	(0.043)	(0.046)
Formal job partcp.	0.174	0.153		
	(0.063)	(0.053)		
Export shock			0.062	0.058
			(0.012)	(0.014)
Observations	1,687	1,687	1,687	1,687
R-squared	0.761	0.772	0.772	0.782

The table reports OLS estimates of veiling and different types of female occupations. At the bottom rows of each panel, formal job participation is replaced instead by the instrument (export shocks). *Veil* is the fraction of female pupils wearing a headscarf, *Formal job participation* is the fraction of female population age 20-24 working in the formal sector, *Informal job partcp.* is the fraction of female population age 20-24 working in informal sector. *family worker* denotes the fraction of female population age 20-24 working as family workers. *Export shock* is the instrument. Each specification in column (2) and (4) also includes the set of controls at the school, district, and national levels. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties and female high-school enrollment rate; National-level control includes economic growth rate. Standard errors for all regressions are clustered at the district level

TABLE 7 – THE EFFECT OF ECONOMIC SHOCKS ON VARIOUS MEASURES OF RELIGIOSITY

	Religiosity		Num. of Prayers		Relig. politics	
	(1)	(2)	(1)	(2)	(1)	(2)
Export shock	0.003	0.003	0.026	0.024	-0.070	-0.072
	(0.011)	(0.011)	(0.041)	(0.039)	(0.051)	(0.051)
Observations	13,654	13,654	12,637	12,637	13,338	13,338
R-squared	0.031	0.071	0.037	0.082	0.109	0.114

Notes: The table reports OLS estimates of three different measures of religiosity on the instrument. The first two column take self-reported religiosity as dependent variable. The value of this variable goes from 1=*not religious* to 4 = *very religious*. The two middle columns take the number of prayers performed in a day as dependent variable; and the last two columns use the importance of religion and religiosity in influencing repondents' choice of political candidates. It takes value 0=*not important at all* to 10=*very important*. The data source is wave 4 and 5 of IFLS (Indonesian Family Life Survey) and the unit of analysis is individuals surveyed in those waves who are resident of the districts sampled in the main analysis of this paper. All regressions includes district and wave fixed effects. Each of the specification numbered (2) also control for age, age-squared, a dummy variable for sex and marriage status. Standard errors for all regressions are clustered at the district level

TABLE 8 – THE REASONS TO VEIL

Reasons	Age group					Total
	16-19	20-24	25-29	30-34	35-39	
Requested by parents	1.10	3.53	2.65	0.00	3.45	2.62
Requested by spouse	0.00	0.32	0.53	2.99	0.00	0.58
Respondent's own will	92.31	86.54	89.95	89.55	93.10	**88.81**
Following trend	0.00	1.28	0.53	0.00	0.00	0.73
Others	6.59	8.33	6.35	7.46	3.45	7.27
Total	100	100	100	100	100	100

Note: The values are in percentage and is sourced from Jakpat Poll on the trend of religious headscarf, conducted in 2016. The question is "What is your reason to wear *Hijab* (a religious headscarf)?" Answer category "Others" include answers such as "Commanded by the Qur'an", "Fear of God's punishment", etc.

TABLE 9 – LOGIT REGRESSION OF VEIL USE ON WORKING STATUS

VARIABLES	Age group			All ages
	< 21	21-30	>30	
	Dep. var: veil(yes=1)			
Base category : Not working				
Cat. 2 : working	-0.94	0.36	0.53	0.30
	(0.46)	(0.14)	(0.20)	(0.11)
Cat. 3: student	0.95	0.37		0.51
	(0.45)	(0.23)		(0.18)
Observations	215	651	130	996

Note: The table shows logistics regression of veiling on working status using the data from Jakpat Poll in 2016. Standard error is clustered at the province level. The unit of analysis is individual respondent. Dependent variable is a dummy valued 1 if wearing a veil. The main variable of interest is working status category. All regression is controlling for age and socio economic status variables.

TABLE 10 – VEILING PREVALENCE ACROSS AGE GROUPS

Wearing a veil?	Age group					Total
	16-19	20-24	25-29	30-34	35-39	
Yes	68.42	72.39	74.41	67.00	67.44	71.59
No	31.58	27.61	25.59	33.00	32.56	28.41
Total	100	100	100	100	100	100

Note: The values are in percentage and is sourced from Jakpat Poll on the trend of religious headscarf, conducted in 2016. The question is "Are you personally wearing *Hijab* (a religious headscarf)?" The respond answers are recoded into Yes when the respond is "yes" and No for other responses

TABLE 11 – VEILING PREVALENCE ACROSS EDUCATION LEVELS

Wearing a veil?	Education Attainment				Total
	Elementary	Secondary 1	Secondary 2	Tertiary	
Yes	64.71	59.42	68.54	76.42	**71.49**
No, but I definitely will	5.88	7.25	9.21	7.86	8.39
No, but maybe I will	17.65	7.25	6.97	8.52	7.89
Sometimes	11.76	21.74	12.58	5.46	9.91
No, and I will not	0.00	4.35	2.70	1.75	2.33
Total	100	100	100	100	100

Note: The values are in percentage and is sourced from Jakpat Poll on the trend of religious headscarf, conducted in 2016. The question is "Are you personally wearing *Hijab* (a religious headscarf)?"

TABLE 12 – VEILING AND FEMALE HIGH SCHOOL ATTENDANCE RATE

			Veil		
	(1)	(2)	(3)	(4)	(5)
Female HS enrollment	0.006	0.018	-0.002	0.021	-0.000
	(0.051)	(0.051)	(0.048)	(0.053)	(0.050)
Formal job partcp.		0.174		0.151	
		(0.061)		(0.052)	
Export shock			0.062		0.058
			(0.012)		(0.014)
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.759	0.772	0.761	0.772	0.782

The table reports OLS estimates of correlation between veiling and female high school attendance rate. *Veil* is the fraction of female pupils wearing a headscarf, *Female HS enrollment rate* is the fraction of high-school age female population attending high school, *Formal job participation* is the fraction of female population age 20-24 working in the formal sector. Each specification in column (4) and (5) also includes the set of controls at the school, district, and national levels. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, and vote share of Islamist parties, National-level control includes economic growth rate. All regressions include district fixeds effect and district-specific time trends. Standard errors for all regressions are clustered at the district level

TABLE 13 – FAMILY INCOME EFFECT

	Veil				
	(1)	(2)	(3)	(4)	(5)
SECOND-STAGE ESTIMATES					
Male job partcp.	1.139	1.134	1.146	1.134	1.139
	(0.629)	(0.621)	(0.715)	(0.680)	(0.800)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.624	0.634	0.633	0.634	0.635
FIRST-STAGE ESTIMATES					
	Male job participation				
Export shock (male)	0.0218	0.0217	0.020	0.0207	0.018
	(0 .005)	(0.005)	(0.005)	(0.005)	(0.005)
K-P Wald F-statistics	20.88	20.77	17.26	18.83	14.29

The table presents 2SLS estimates corresponding to the results of table 2, but using male employment as endogenous variable. The instrument is constructed using similar components of the main instrument, except for the female score index, which is replaced by male score index. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of female population age 20-24 who are working in the formal sector, vote share of Islamist parties and female high-school enrollment rate, National-level control includes economic growth rate. All specifications include district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level

TABLE 14 – MEANS OF TRANSPORTATION AND DISTANCE TO WORK

	Formal job=1					
	(1)	(2)	(3)	(4)	(5)	(6)
Base cat.		*On foot*			*Less than 10 KM*	
Some vehicles	0.130	0.080	0.100			
	(0.002)	(0.002)	(0.002)			
10-29 KM				0.063	0.023	0.028
				(0.000)	(0.000)	(0.000)
$\geq$ *30 KM*				0.046	0.006	0.022
				(0.000)	(0.000)	(0.000)
District FE		✓	✓		✓	✓
Controls			✓			✓
Observations	6,729,176	6,729,176	6,729,176	6,510,228	6,510,228	6,510,228
R-squared	0.001	0.116	0.156	0.005	0.114	0.154

The first three columns reports simple regressions of the probability of having a formal job on the dummy of means of transportation to work, while the last three reports simple regressions of the probability of having a formal job on the dummy for distance to work. The outcome variable values 1 if the job is formal and zero if the job is informal, defined as per definition used in text. The base category for distance to work is less than 10 KM, while the base category for means of transportation is going to work on foot or walking. The vehicles used are either public, private, or shared. The unit of analysis is individuals surveyed in Sakernas 2014 and declare to be working. In columns (3) and (6) we also control for dummy for gender and age groups.

TABLE 15 – HETEROGENEITY EFFECT BY VEILING FRACTION AT T-1

	$Veil_t$				
Export shock	0.032	0.031	0.027	0.031	0.027
	(0.008)	(0.008)	(0.009)	(0.008)	(0.009)
$Veil_{t-1}$	0.586	0.578	0.570	0.578	0.570
	(0.093)	(0.092)	(0.091)	(0.092)	(0.091)
Export shock$*Veil_{t-1}$	-0.016	-0.016	-0.013	-0.016	-0.013
	(0.016)	(0.016)	(0.016)	(0.016)	(0.015)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,434	1,434	1,434	1,434	1,434
R-squared	0.840	0.848	0.849	0.848	0.849

The table reports reduced-form estimates with interaction following equation 6. $Veil_t$ is the fraction of female pupils wearing a headscarf at time t, $Veil_{t-1}$ is the fraction of female pupils wearing a headscarf at time $t-1$, *Export shock* is the instrument. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties and female high-school enrollment rate, National-level control includes economic growth rate. Standard errors for all regressions are clustered at the district level

B Robustness checks

TABLE 16 – ROBUSTNESS: TRIMMING OUTLIERS

	Veil				
	(1)	(2)	(3)	(4)	(5)
Formal job participation	2.492	2.515	2.922	2.523	2.934
	(0.664)	(0.689)	(0.879)	(0.709)	(0.937)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,643	1,643	1,643	1,643	1,643
R-squared	0.269	0.263	0.154	0.260	0.148

The table presents 2SLS estimates corresponding to the results in Panel B of table 2, but excluding observations with 100 percent female pupils with headscarves. Variable definitions as well as the set of controls are similar to those presented in the corresponding table 2.

A Alternative IV with global import

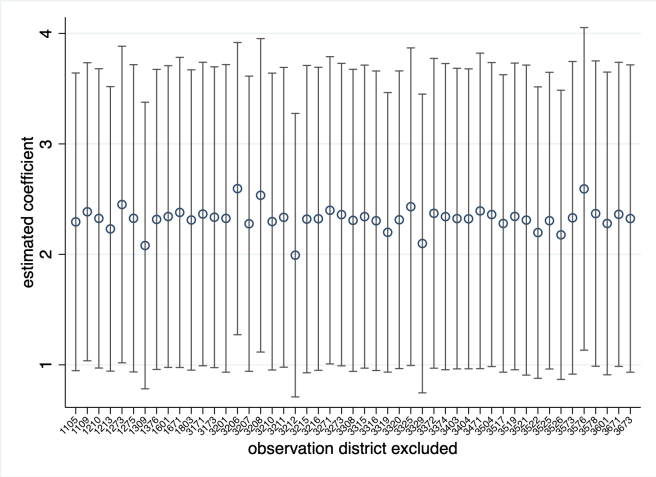
This alternative instrument is still taking the fundamental logic of the original instrument, but instead of using the value of Indonesia's export, we use the global demand for commodities, i.e. the total value of global import from all other countries but Indonesia. It is computed following the formula below:

$$global_demand_shock_{dt} = \sum_{k=1}^{K} global_import_{kt} \underbrace{\frac{L_{kd,t=1993}}{L_{d,t=1993}}}_{sector_share_{kd}} \underbrace{\frac{L_{fk,t=1993}}{L_{k,t=1993}}}_{\text{female score_k}} \quad (7)$$

$$global_import_{kt} = \sum_{n=1}^{N} import_{knt} - import_{kt_Indo} \quad (8)$$

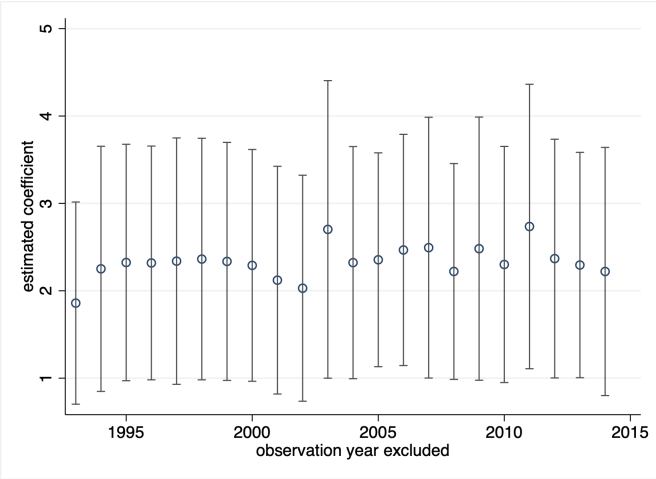
Where $\sum_{n=1}^{N} import_{knt}$ is the sum of global import of commodity k at time t by all countries; $import_{kt_Indo}$ is the value of import of commodity k by Indonesia at time t; and N is the number of all countries importing commodity k at time t.

FIGURE 3 – Jackknife resampling estimates by district



(a) *Note*: This figure displays the re-estimation of the coefficient (within 95 percent confidence interval) in the main regression presented in table 2–model 1, using a jackknife resampling method over district dimension.

FIGURE 4 – Jackknife resampling estimates by year



(a) *Note*: This figure displays the re-estimation of the coefficient (within 95 percent confidence interval) in the main regression presented in table 2–model 1, using a jackknife resampling method over year dimension.

TABLE 17 – ROBUSTNESS: TRIMMING OBSERVATIONS WITH HIGH VARIANCE

	Veil				
	(1)	(2)	(3)	(4)	(5)
Formal job participation	2.113	2.108	2.426	2.101	2.414
	(0.642)	(0.637)	(0.800)	(0.660)	(0.852)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,601	1,601	1,601	1,601	1,601
R-squared	0.505	0.516	0.461	0.517	0.465

The table presents 2SLS estimates corresponding to the results in Panel B of table 2, but excluding observations from district with only one school in the sample. All definitions of variable as well as the set of controls are similar to the corresponding table 2

TABLE 18 – ALTERNATIVE IV WITH GLOBAL IMPORT

Panel A. REDUCED FORMS ESTIMATES					
	Veil				
Global import shock	0.094	0.095	0.089	0.091	0.079
	(0.029)	(0.029)	(0.040)	(0.028)	(0.042)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.762	0.771	0.772	0.772	0.773
Panel B. 2SLS ESTIMATES					
	Veil				
Formal job partcp.	2.932	2.963	2.865	3.006	2.907
	(1.194)	(1.204)	(1.394)	(1.294)	(1.709)
Panel C. FIRST STAGE ESTIMATES					
	Veil				
Global import shock	0.032	0.032	0.031	0.030	0.027
	(0.012)	(0.012)	(0.014)	(0.012)	(0.015)
K-P Wald F-Statistics	7.27	7.14	4.85	6.09	3.13

The table reports the reduced forms and 2SLS estimates using an alternative instrument, *Global import shock*. Unlike the original instrument, it takes the value of world import (minus Indonesia's import) rather than the value of commodity exported by Indonesia's producers. *Veil* is the fraction of female pupils wearing veil, *Formal job participation* is the fraction of female population age 20-24 who works in the formal sector. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties and female high-school enrollment rate, National-level control includes economic growth rate. Panel B and C also include the same set of additional controls (School, District, and National levels) as Panel A. All specifications include district fixed effects and district-specific time trends. Standard errors for all regressions are clustered at the district level

TABLE 19 – MAIN RESULTS USING LAGS OF ECONOMIC SHOCKS

Reduced Forms Estimates

	$Veil_{sdt}$					
Export shock$_{d,t-1}$	0.056	0.054				
	(0.014)	(0.015)				
Export shock$_{d,t-2}$			0.043	0.041		
			(0.014)	(0.014)		
Export shock$_{d,t-3}$					0.039	0.036
					(0.014)	(0.013)
School-level controls		✓		✓		✓
District-level controls		✓		✓		✓
National-level controls		✓		✓		✓
Observations	1,610	1,610	1,529	1,529	1,529	1,529
R-squared	0.775	0.785	0.784	0.793	0.782	0.791

2SLS Estimates

	$Veil_{sdt}$					
Formal job partcp$_{d,t-1}$	1.998	1.965				
	(0.647)	(0.680)				
Formal job partcp$_{d,t-2}$			1.363	1.319		
			(0.532)	(0.521)		
Formal job partcp$_{d,t-3}$					1.419	1.415
					(0.567)	(0.566)
School-level controls		✓		✓		✓
District-level controls		✓		✓		✓
National-level controls		✓		✓		✓
Observations	1,610	1,610	1,529	1,529	1,529	1,529
R-squared	0.551	0.572	0.701	0.718	0.681	0.690

The table presents reduced forms and 2SLS estimates of veiling on different lags of economic shocks. The unit of analysis is school-district-year. *Veil* denotes the fraction of female students who wear a veil, *Export shock* denotes the instrument, *Formal job participation* denotes the fraction of female population age 20-24 who works in the formal sector. School-level controls include male to female student ratio, the size of student body, the latitute and the longitude of school location; District-level controls include fraction of population who live in urban areas, the fraction of male population age 20-24 who are working, vote share of Islamist parties, and female high-school enrollment rate; National-level control includes economic growth rate. All specifications include district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level

TABLE 20 – MAIN RESULTS USING DISTRICT-SPECIFIC-SHARE FEMALE SCORE INDEX

	Veil				
	(1)	(2)	(3)	(4)	(5)
Formal job partcp.	2.108	2.103	2.502	2.095	2.494
	(0.633)	(0.631)	(0.894)	(0.649)	(0.952)
School-level controls		✓	✓	✓	✓
District-level controls			✓		✓
National-level controls				✓	✓
Observations	1,687	1,687	1,687	1,687	1,687
R-squared	0.498	0.508	0.428	0.511	0.430

The table presents 2SLS estimates corresponding to the results in Panel B of table 2, but using a slightly different construction of the instrument. The instrument in this table is constructed using district-specific female score index, while the instrument in the main text is using a national female score index that are applied homogenously to all districts in the sample. Variable definitions as well as the set of controls are similar to those presented in the corresponding table 2. All specifications include district fixed effects and district specific time trends. Standard errors for all regressions are clustered at the district level

C Variable definition, calculation and data source

A International trade data

The first component of the instrument, the value of of Indonesia's export (alternatively, the value of global import), is sourced from UN-COMTRADE database, **The World Integrated Trade Solution (WITS)** at https://wits.worldbank.org/ in July 2018. The interface allows users to customize thire query, e.g. choosing the reporting country, trade partner, the year of report, as well as the nomenclature of product classification. The nomenclature used in this paper is ISIC Revision 2 at 3 digits level, which could be matched to the industrial classification of Statistics Indonesia for Large and Medium Industry census in 1993.

The trading values acquired is then transformed into its constant value in 1990 USD using GDP deflator for the United States downloaded from **Penn World Table-international comparisons of production, income and prices 9.1** at https://febpwt.webhosting.rug.nl/ accessed in August 2018.

After deflated so that the value are comparable across years, these numbers are then normalized to 1 in 1990 so that they are comparable across different industrial sector. This way, what we essentially capturing is the normalized real value of the demand for specific product across different time period.

B District sectoral composition and industrial female scores

The second and the third component of *Export shock* are the female score index of the industry and district Industrial composition. They are calculated using the Medium and Large Manufacturing census produced annually by Statistics Indonesia (Badan Pusat Statistik-BPS). The census provides information about the location, output/product classification, and since 1993, the breakdown of male and female labor used in the production of the firm.

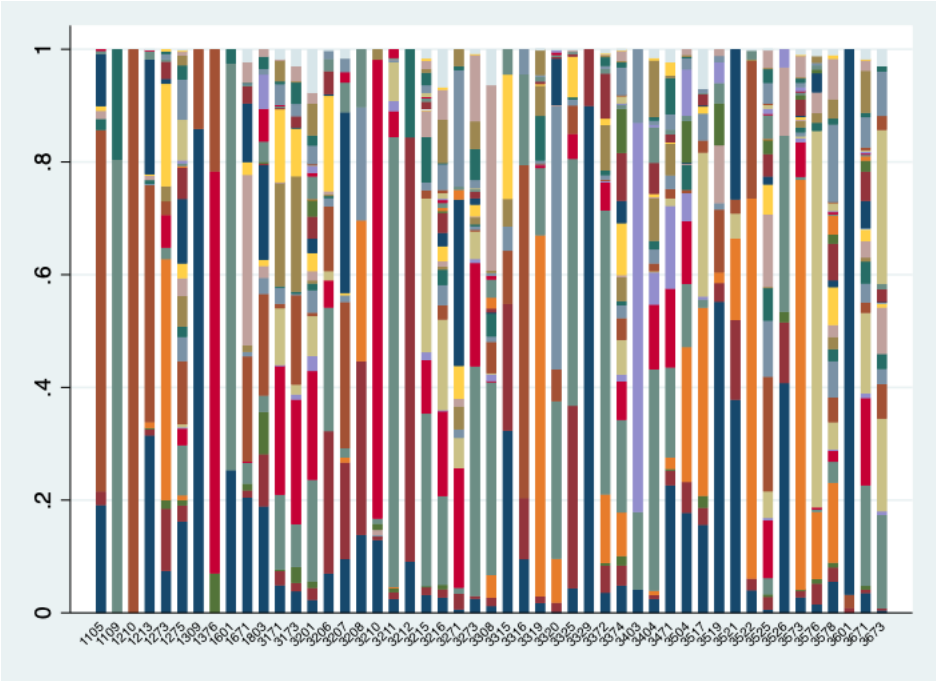


FIGURE 5 – HISTORICAL INDUSTRIAL COMPOSITION BY DISTRICT

Note: The figure shows the industrical composition in each district (district code in x axis) at the base year (1993) disaggregated by 3 digit ISIC Rev.2 industrical classification. Each color represent one industrial sector

The historical sectoral composition is calculated as the base-year ratio of labor force who work in a given sector in a given district and the total worker of of manufacturing sector in the district. The base year used is 1993. In figure 5, one could observe the variation in historical industrial composition for districts included in the sample.

While **industrial female score** is an index calculated as the ratio of female worker and the total worker in a given industry at the national level. This is also calculated at the base year, ie. the year 1993. These scores are presented in table 22

TABLE 21 – OPERATIONALIZATION OF VARIABLES

VARIABLES	Details	Source
Veil	Fraction of female pupils who wear a headscarf in school book pictures in a given school-district-year*	School book, see details in part D below
Formal job partcp.	Fraction of female population aged 20-24 who hold a formal job. Holding a formal job is defined as satistifying two criteria: 1. Working is the activities taking most time in the previous week. 2.Occupation status is employee = code 4 in SAKERNAS coding	SAKERNAS, various years (1993-2014)*
Export Shock	Demand shocks for the product produced by Indonesian based firms in the international market	See text section B and appendix C below for detailed calculation.
Informal job partcp.	Share of female aged 20-24 who hold an informal job. An informal job is defined as satisfying two criteria: 1. Working is the activities taking most time in the previous week. 2. Occupation status do not fall into category 4 (employee). This includes independent workers (with or without employees), casual workers (in agriculture or non agricultural sector) and (unpaid) family workers	SAKERNAS, various years (1993-2014)*
Family worker	Share of female aged 20-24 who work as a family worker (paid or unpaid). A family worker is defined as satisfying two criteria: 1. Working is the activities taking most time in the previous week. 2. Occupation status as family/unpaid workers= code 5 in SAKERNAS 2001 and earlier; code 7 in SAKERNAS 2002 and later	SAKERNAS, various years (1993-2014)**
Male job partcp.	Fraction of male population aged 20-24 who hold a job. This is defined as declaring working as the main activity in the previous week	SAKERNAS, various years (1993-2014)*
Fraction urban	Fraction of population aged 10 and above who live in urban areas	SAKERNAS, various years (1993-2014)**
Male/female ratio	The ratio of male students to female students in a given school-district-year	School book
Num. of student	The total number of students in a given school-district-year	School book
Latitude, Longitude	The geographic coordinate of school location	http://sekolah.data.kemdikbud.go.id/
Islamist vote	Vote shares of Islamist party (PPP, PKS, PBB)*** at the electoral college level	KPU (Election Comission)
Economic growth	Economic growth at the national level	The World Bank
Female HS enrollment	Fraction of female population aged 15-19 years old whose main activities is going to school in the previous week	SAKERNAS, various years (1993-2014)**
Female score	The share of female labor in sector k in the base year,1993	Medium and Large Manufacturing Census (Statistik Industri)
Sector share	The share of sector k in district d in the base year,1993	Medium and Large Manufacturing Census (Statistik Industri)
Value of export	The value of Indonesia's export in sector k in year t	UN Comtrade

Note: *We interpolate the missing values in this variable. This procedure does not change the mean or the standard deviation up to 2 decimal point.
**SAKERNAS 1995 is not available, therefore for all variables sourced from SAKERNAS, we interpolate the value for this year.
***only PPP before 1999 election.

TABLE 22 – FEMALE SCORE INDEX (3 DIGIT ISIC REV.2)

Code	Description	Female Share
371	Iron and steel basic industries	.053
369	Manufacture of other non-metallic mineral products	.099
363	Manufacture of other non-metallic mineral products	.100
354	Manufacture of miscellaneous products of petroleum and coal	.108
384	Manufacture of transport equipment	.119
382	Manufacture of machinery except electrical	.124
351	Manufacture of industrial chemicals	.198
372	Non-ferrous metal basic industries	.216
381	Manufacture of fabricated metal products, except machinery and equipment	.238
341	Manufacture of paper and paper products	.239
362	Manufacture of glass and glass products	.240
355	Manufacture of rubber products	.249
342	Printing, publishing and allied industries	.309
332	Manufacture of furniture and fixtures, except primarily of metal	.322
361	Manufacture of pottery, china and earthenware	.373
331	Manufacture of wood and wood and cork products, except furniture	.374
313	Beverage industries	.374
311	Food manufacturing	.378
323	Manufacture of leather and products of leather, leather substitutes and fur, except footwear and wearing apparel	.378
312	Food manufacturing	.423
364	Manufacture of other non-metallic mineral products	.464
385	Manufacture of professional and scientific, and measuring and controlling equipment not elsewhere classified, and of photographic and optical goods	.480
356	Manufacture of plastic products not elsewhere classified	.506
352	Manufacture of other chemical products	.510
383	Manufacture of electrical machinery apparatus, appliances and supplies	.519
321	Manufacture of textiles	.531
390	Other Manufacturing Industries	.686
324	Manufacture of footwear, except vulcanized or moulded rubber or plastic footwear	.739
322	Manufacture of wearing apparel, except footwear	.763
314	Tobacco manufactures	.793

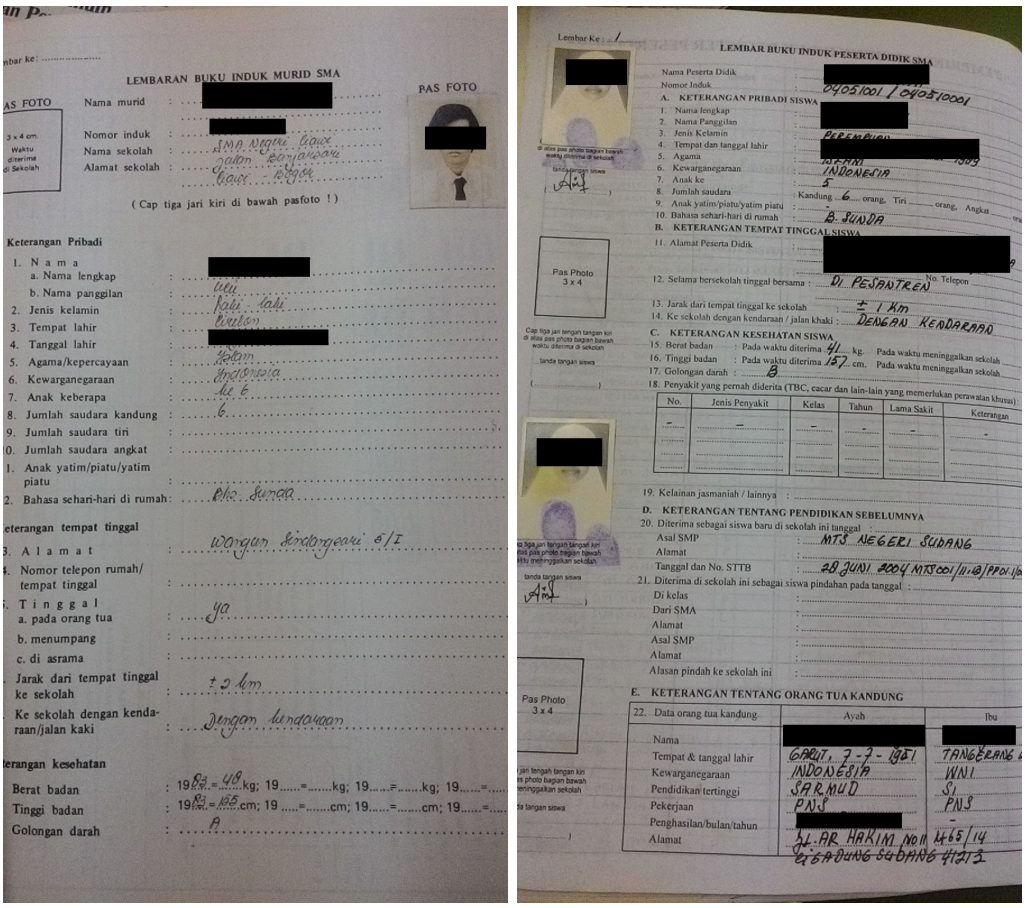


FIGURE 6 – SAMPLES OF HIGH SCHOOL BOOK PAGES

D Veil : Data collection procedure

The data on veiling rate is obtained by calculating the number of female pupils wearing a headscarf in the high-school book pictures as a proportion of all female pupils. This book is a register, a document normally kept by administrative office of public schools in Indonesia. Examples of pages from this document is presented in figure 6

A Sampling

The sample frame is all districts (total number is 267) located in Java and Sumatera (two main islands in Indonesia). The data collection is performed between March - August 2017. The number of districts included in the original sample is 50. They are randomly chosen with the number pro-

portional to the size of population inhabited the province in 2014, a choice taken to maximise the efficiency of data collection. In table 23, one could observe the characteristics of non-sampling and sampling districts. Apart from the number of population, there is no significant differences between the two group. From each of the district chosen in the sample, two schools are randomly selected. Out of 50 districts originally planned, 1 district dropped due to logistical problem, no reliable enumerator is found to perform data collection in this district. Out of 49 district successfully sampled, 5 districts only resulting in single school. The geographic distribution the district sample could be observed in figure 1

TABLE 23 – DISTRICTS CHARACTERISTICS: SAMPLING VS. NON-SAMPLING

Variables (2014)	Java		Sumatra	
	Non-sample mean	Difference	Non-sample mean	Difference
Development indicators				
Access to electricity	99.744	0.072	94.966	2.318
		(0.068)		(2.416)
Access to safe sanitation	71.722	1.980	71.532	1.433
		(2.438)		1.433
Percapita expenditure (log)	13.432	0.094	13.446	0.048
		(0.065)		(0.085)
Percapita expenditure–poorest 20%(log)	12.567	0.051	12.693	0.041
		(0.046)		(0.070)
Birth attended by professional (%)	92.447	-0.312	88.662	4.573
		(2.039)		(3.350)
Poverty rate	11.446	-1.100		-0.138
		(0.972)	11.984	(1.835)
Education indicators				
Human Development Index	68.961	1.712	67.304	2.445
		(1.102)		(1.491)
Literacy rate	94.036	1.213	97.509	0.964
		(0.922)		(0.990)
Enrollment rate: primary school	96.763	0.589	97.122	-0.097
		(0.403)		(0.656)
Enrollment rate : junior secondary	80.091	1.862	78.061	3.091
		(1.308)		(2.169)
Enrollment rate : senior secondary	60.308	0.558	64.592	-1.230
		(1.878)		(2.897)
Population and Labor market				
Labor force participation rate	0.510	(0.002)	0.471	-0.005
		(0.010)		(0.019)
Employment rate	0.481	-0.004	0.446	-0.013
		(0.011)		(0.021)
Unemployment rate	0.029	0.002	0.025	0.008*
		(0.002)		(0.004)
Underemployment rate	0.140	0.000	0.177	-0.022
		(0.013)		(0.022)
Population (million)	1.086	0.283	0.329	0.379
		(0.162)		(0.097)
Morbidity rate (percent)	30.695	1.422	27.095	-0.923
		(1.422)		(2.319)
Local governance indicators				
Local finance audit score	2.744	0.173	2.162	0.656
		(0.318)		(0.426)
Government transfer – DAK	24.637	0.043	24.538	0.041
		(0.176)		(0.217)

Note: The table shows the difference in various outcome variables between sample and non-sample districts (sample values - non sample values) in each of island, Java and Sumatra. Data source: Indo-DAPOER, the World Bank, Jakarta; analysis by the author. Standard errors in parenthesis.

TABLE 24 – List of sample districts

No	District code	District name	Province name	# schools sampled
1	1105	Aceh Timur, Kab.	Aceh	2
2	1109	Pidie, Kab.	Aceh	2
3	1210	Dairi, Kab.	Sumatera Utara	2
4	1213	Langkat, Kab.	Sumatera Utara	2
5	1273	Pematang Siantar, Kota	Sumatera Utara	2
6	1275	Medan, Kota	Sumatera Utara	2
7	1309	Pasaman, Kab.	Sumatera Barat	2
8	1376	Payakumbuh, Kota	Sumatera Barat	2
9	1471	Pekanbaru, Kota	Riau	dropped
10	1601	Ogan Komering Ulu, Kab.	Sumatera Selatan	2
11	1671	Palembang, Kota	Sumatera Selatan	2
12	1803	Lampung Selatan, Kab.	Lampung	1
13	3171	Jakarta Selatan, Kota	DKI Jakarta	2
14	3173	Jakarta Pusat.	DKI Jakarta	1
15	3201	Bogor, Kab.	Jawa Barat	2
16	3206	Tasikmalaya, Kab.	Jawa Barat	2
17	3207	Ciamis, Kab.	Jawa Barat	1
18	3208	Kuningan, Kab.	Jawa Barat	2
19	3210	Majalengka, Kab.	Jawa Barat	2
20	3211	Sumedang, Kab.	Jawa Barat	2
21	3212	Indramayu, Kab.	Jawa Barat	2
22	3215	Karawang, Kab.	Jawa Barat	2
23	3216	Bekasi, Kab.	Jawa Barat	2
24	3271	Bogor, Kota	Jawa Barat	2
25	3273	Bandung, Kota	Jawa Barat	2
26	3308	Magelang, Kab.	Jawa Tengah	2
27	3315	Grobogan, Kab.	Jawa Tengah	2
28	3316	Blora, Kab.	Jawa Tengah	2
29	3319	Kudus, Kab.	Jawa Tengah	1
30	3320	Jepara, Kab.	Jawa Tengah	2
31	3325	Batang, Kab.	Jawa Tengah	2
32	3329	Brebes, Kab.	Jawa Tengah	2
33	3372	Surakarta, Kota	Jawa Tengah	2
34	3374	Semarang, Kota	Jawa Tengah	2
35	3403	Gunung Kidul, Kab.	D.I. Yogyakarta	2
36	3404	Sleman, Kab.	D.I. Yogyakarta	2
37	3471	Yogyakarta, Kota	D.I. Yogyakarta	2
38	3504	Tulungagung, Kab.	Jawa Timur	2
39	3517	Jombang, Kab.	Jawa Timur	1
40	3519	Madiun, Kab.	Jawa Timur	2
41	3521	Ngawi, Kab.	Jawa Timur	2
42	3522	Bojonegoro, Kab.	Jawa Timur	2
43	3525	Gresik, Kab.	Jawa Timur	2
44	3526	Bangkalan, Kab.	Jawa Timur	2
45	3573	Malang, Kota	Jawa Timur	2
46	3576	Mojokerto, Kota	Jawa Timur	2
47	3578	Surabaya, Kota	Jawa Timur	2
48	3601	Pandeglang, Kab.	Banten	2
49	3671	Tangerang, Kota	Banten	2
50	3673	Serang, Kab.	Banten	2

B Data collection procedure

Enumerators are given a (randomly ordered) list of schools and visit the schools based on the order. The sample frame of this study does not include Islamic schools (public Madrasahs). This is done in purpose because all female pupils in Madrasahs wear a headscarf, hence there is no information to obtain from these religious schools.

The instruction of data collection procedure at the school level is as follows:

1. Obtain permission from school administration to collect data by showing the necessary documents from the PI and the government authorities.

2. Check if the quality of register book pass the standard (readable, fairly organized and available for many cohort years). If the quality of material below standard, go to the next school in the list, otherwise, go to the next step.

3. Count the number of students in each cohort, male and female and register them in the data collection sheets provided by PI.

4. Count the number of female students who wear a headscarf and register them in the data collection sheets provided by PI.

5. Scan/take pictures of randomly selected pages of the register book. Enumerators take scans of every other page of the available books, to ease the work, PI set the rules that odd pages are to be taken for odd year cohorts and even pages are to be taken for even year cohort.

6. Upload all information collected and the pictures/scan in the dedicated online server.

7. Data quality control by PI and team, approval if quality is accepted and the scans matched the information in the data collection sheets.

8. Amend or collect the missing information/data if necessary.

9. Repeat the procedure for the next school.